# COURSE BOOK B.TECH. III YEAR

## (Autonomous)

## Computer Science and Engineering -
## Artificial Intelligence & Machine Learning (AIML)

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# CURRICULUM STRUCTURE & SYLLABUS

**Effective from the Session: 2026-27**




## VISION

To become a leading institution nationally in the area of professional education, research & innovation for serving the global community.

## MISSION

- To impart quality professional education, skills and values through outcome-based innovative teaching learning process in all spheres.
- To undertake collaborative interdisciplinary research as a co-requisite for professional education and simultaneously solve problems faced by society and industry.
- To create an ambience of innovation, entrepreneurship and consultancy for future leaders and innovators.
- To keep faculty members enthusiastic by continuous professional development and positive working environment.

# Department of Computer Science & Engineering (AIML)

## VISION

To emerge as a globally competent leader by fostering industry-linked, innovative learning and impactful research in Computational Intelligence.

## MISSION

- To develop students with strong foundation of computer science with focus on Artificial Intelligence and other emerging technologies through outcome-based teaching learning process.
- To collaborate with industry for skill enhancement of teaching professionals and students emphasizing on project-based learning.
- To prepare ethically strong students with powerful leadership skills.

## B.Tech. Computer Science & Engineering (AIML)

### PROGRAM EDUCATIONAL OBJECTIVE

**Graduates will be able to:**

- Be employable in the field of Computing and Artificial Intelligence for solving real life problems.
- Be able to adapt with ever changing technological advancements in dynamic professional world.
- Be able to make a successful career in research or higher studies.

### PROGRAM OUTCOMES

**Engineering Graduates will be able to:**

**PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

**PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

**PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

**PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

**PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

**PO6: The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

**PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

**PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

**PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

**PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

**PO11: Life-Long Learning:** Recognize the need for, and have the preparation and ability for

i) independent and life-long learning
ii) adaptability to new and emerging technologies and
iii) critical thinking in the broadest context of technological change. (WK8)

## CORE VALUES

- Academic excellence
- Collaborative and interdisciplinary research culture
- Conducive eco-system
- Strong humanitarian values & Ethics

## B.Tech CSE(AIML) 5th Sem

| S No. | Course Category (AICTE) | Course Category (UGC) | BOS | Course Code | Course Name | Type | Academic Learning (AL) | | | Continuous Internal Examination (CIE) | | | End Sem Examination (ESE) | Total Marks | Total Credits |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | L | T | P | MSE | CA | TOTAL | | CIE+ ESE | |
| 1 | ES | Major (Core) | IT | IT302L | Computer Networks | L | 3 | 0 | 0 | 60 | 15 | 75 | 75 | 150 | 3 |
| 2 | HS | AEC | ASH | HS203L | Aptitude - 3 | L | 1 | 0 | 0 | - | 25 | 25 | 25 | 50 | 1 |
| 3 | HS | AEC | ASH | HS204L | Soft Skills for Employability - 1 | L | 2 | 0 | 0 | - | 100 | 100 | - | 100 | 2 |
| **Blended** | | | | | | | | | | | | | | | |
| 4 | PC | Major (Core) | CSIT | IT401B | Advance Databases | B | 3 | 0 | 2 | 80 | 20 | 100 | 100 | 200 | 4 |
| 5 | PC | Major (Core) | CSE(AI) | AI401B | NLP and Large Language Models | B | 2 | 0 | 2 | 60 | 15 | 75 | 75 | 150 | 3 |
| 6 | PC | Major (Core) | CSE(AIML) | AI316B | Software System Engineering and DevOps | B | 2 | 0 | 2 | 60 | 15 | 75 | 75 | 150 | 3 |
| 7 | PE | Major (Core) | - | - | Professional Elective – III | B | 3 | 0 | 2 | 80 | 20 | 100 | 100 | 200 | 4 |
| **Lab/Practical** | | | | | | | | | | | | | | | |
| 8 | ES | Major (Core) | IT | IT302P | Computer Networks Lab | P | 0 | 0 | 2 | - | 25 | 25 | 25 | 50 | 1 |
| 9 | PC | SEC | CS | CS313P | Problem Solving Lab-1 | P | 0 | 0 | 4 | - | 100 | 100 | - | 100 | 2 |
| 10 | PW | Research Project / Dissertation | CSIT | IT304P | Project Internship | P | 0 | 0 | 0 | - | 50 | 50 | - | 50 | 1 |
| **Total Hours : 30 hrs.** | | | | | | | **16** | **0** | **14** | | | | | **1200** | **24** |

## B.Tech CSE(AIML) 6th Sem

| S No. | Course Category (AICTE) | Course Category (UGC) | BOS | Course Code | Course Name | Type | Academic Learning (AL) | | | Continuous Internal Examination (CIE) | | | End Sem Examination (ESE) | Total Marks | Total Credits |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | L | T | P | MSE | CA | TOTAL | | CIE+ ESE | |
| 1 | PC | Major (Core) | CSE(AIML) | AI402L | Automata Theory and Compiler Design | L | 3 | 0 | 0 | 60 | 15 | 75 | 75 | 150 | 3 |
| 2 | HS | AEC | ASH | HS205L | Aptitude - 4 | L | 1 | 0 | 0 | - | 25 | 25 | 25 | 50 | 1 |
| 3 | OE | Multidisciplinary | - | - | Open Elective-I | L | 2 | 0 | 0 | 40 | 10 | 50 | 50 | 100 | 2 |
| **Blended** | | | | | | | | | | | | | | | |
| 4 | PC | Major (Core) | CSE(AI) | AI202B | Fundamentals of Cyber Security | B | 3 | 0 | 2 | 80 | 20 | 100 | 100 | 200 | 4 |
| 5 | PC | Major (Core) | CSE(AIML) | AI403B | System Design | B | 3 | 0 | 2 | 80 | 20 | 100 | 100 | 200 | 4 |
| 6 | PE | Major (Core) | - | - | Professional Elective – IV | B | 1 | 0 | 2 | 40 | 10 | 50 | 50 | 100 | 2 |
| **Lab/Practical** | | | | | | | | | | | | | | | |
| 7 | HS | AEC | ASH | HS206P | Soft Skills for Employability - 2 | P | 0 | 0 | 2 | - | 50 | 50 | - | 50 | 1 |
| 8 | PC | SEC | CS | CS317P | Problem Solving Lab-2 | P | 0 | 0 | 2 | - | 50 | 50 | - | 50 | 1 |
| 9 | PW | Research Project / Dissertation | CSIT | IT305P | Minor Project Work | P | 0 | 0 | 6 | - | 75 | 75 | 75 | 150 | 3 |
| **Total Hours : 29 hrs.** | | | | | | | **13** | **0** | **16** | | | | | **1050** | **21** |

## Professional Electives (PE) – Offered to CSE(AIML) Students

| Course Type (PE) | Artificial Intelligence and Machine Learning | Cloud Ops and Security | Data Engineering |
|---|---|---|---|
| **BOS** | **CSE(AIML)** | **CSE(AI)** | **CSE(AIML)** |
| **PE-III (5thSem)** | Agentic AI (AI405E) | Cloud Security Foundations (AI406E) | Advanced Analytics & Visualization (AI407E) |
| **PE-IV (6thSem)** | Quantum Computing Essentials (AI312E) | Quantum Computing Essentials (AI312E) | Quantum Computing Essentials (AI312E) |
| | Ethical AI (AI313E) | Ethical AI (AI313E) | Ethical AI (AI313E) |
| | Mobile Application Development (AI314E) | Mobile Application Development (AI314E) | Mobile Application Development (AI314E) |
| | System Testing (AI315E) | System Testing (AI315E) | System Testing (AI315E) |

## Open Electives -Offered to CSE(AIML) Students

| BOS | ASH | ECE | EEE | ME | KSOM |
|---|---|---|---|---|---|
| **Open Elective-I (6th Sem)** | Reliability Theory & Maintenance (MA202O) | Introduction to Robotic Systems (EC237O) | AIoT for Smart Energy (EE214O) | Advanced Engineering Materials (ME229O) | Business Strategy (MS203O) |

# Theory Courses Detail Syllabus – V Sem

| Course Code: IT302L | Course Name: Computer Networks | L | T | P | C |
|---|---|---|---|---|---|
| | | 3 | 0 | 0 | 3 |

**Pre-requisite: Computer Fundamentals**

**Course Objectives:**

The objective of this course is to provide insight about computer network concepts and to gain comprehensive knowledge of the layered communication architectures, its functionalities, key protocols as well as security issues related with each layer.

**Course Outcome:** After completion of the course, the student will be able to

1. Apply the knowledge of networking concepts and functionality of physical layer.
2. Explore the concept of elementary data link layer protocol to build a robust network.
3. Analyze the concept of routing and IP addressing in network layer.
4. Examine the usage and working of transport layer.
5. Determine the performance of different protocols used at application layer.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | - | - | - | - | - | - | - | - | 2 |
| CO2 | 3 | 3 | - | - | 2 | - | - | - | - | - | 2 |
| CO3 | 3 | 3 | - | - | 2 | - | - | - | - | 2 | 2 |
| CO4 | 3 | 3 | - | - | - | - | - | - | - | - | 2 |
| CO5 | 3 | 3 | - | - | - | - | - | - | - | - | 2 |

| Unit 1 | Introduction -Physical Layer | 9 hours |
|---|---|---|

**Introduction to computer networks:** Network applications, Mode of communications, LAN, MAN, WAN, Internet, network hardware, network software, Design issues of layers, reference models: OSI, TCP/IP layers, and characteristics.
**Physical Layer:** Network devices, Network topology, Transmission media, Signal transmission, Network performance and transmission impairments, the public switched telephone network, Switching techniques.

| Unit 2 | Data Link Layer | 9 hours |
|---|---|---|

**Data Link layer:** Design issues, Framing, Error Detection and Correction, Flow control (Elementary Data Link Protocols, Sliding Window protocols).
**Medium Access Control and Local Area Networks:** Multiple access protocols: Random Access (ALOHA, CSMA, CSMA/CD, CSMA/CA), Controlled Access (Polling, Reservation, Token Passing), Channelization (FDMA, TDMA, CDMA). IEEE standards Wired LANs (Ethernet): Traditional Ethernet, Fast Ethernet, Gigabit Ethernet, Wireless LANs: IEEE 802.11 and Bluetooth.

| Unit 3 | Network Layer | 9 hours |
|---|---|---|

**Network Layer:** Network layer design issues, Logical addressing: Basic internetworking, Subnetting, Supernetting, NAT, network layer protocols, IPv4, IPv6
**Routing:** Static and dynamic routing, Routing algorithms and protocols (Shortest Path Algorithms, Flooding, Distance Vector Routing, Link State Routing, Hierarchical Routing, Broadcast Routing, Multicast Routing, Anycast Routing: RIP, OSPF, BGP)

| Unit 4 | Transport Layer | 9 hours |
|---|---|---|

**Transport Layer :** Process-to-process delivery, Transport layer protocols (UDP and TCP), SCTP, Connection management,
**Congestion control in TCP:** Congestion Control: Open Loop, Closed Loop choke packets; Quality of service: techniques to improve QoS: Leaky bucket algorithm, Token bucket algorithm

| Unit 5 | Application Layer | 9 hours |
|---|---|---|

**Session Layer:** Roles and responsibility of session layer protocols
**Presentation Layer: Cryptography** Symmetric-key cryptography, Asymmetric-key cryptography, Digital Signature,
**Compression:** Lossless and Lossy Compression

**Application Layer:** Domain Name System, World Wide Web and Hyper Text Transfer Protocol, Electronic mail, File Transfer Protocol, Simple Network Management Protocol, Telnet.

| Total Lecture Hours | 45 hours |
|---|---|

**Textbook:**

**1.** Behrouz Forouzan, "Data Communication and Networking" Fourth Edition-2006, Tata McGraw Hill
**2.** Andrew Tanenbaum "Computer Networks", Fifth Edition-2011, Prentice Hall.
**3.** William Stallings, "Data and Computer Communication", Eighth Edition-2008, Pearson.

**Reference Books:**

**1.** Kurose and Ross, "Computer Networking- A Top-Down Approach", Eighth Edition-2021, Pearson.
**2.** Peterson and Davie, "Computer Networks: A Systems Approach", Fourth Edition-1996, Morgan Kaufmann
**3.** Behrouz Forouzan, "TCP/IP Protocol Suite", McGraw Hill.

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| MSE | CA | ESE | Total Marks |
| 60 | 15 | 75 | 150 |
| 75 | | | |

| Course Code: HS203L | Course Name: Aptitude - 3 | L | T | P | C |
|---|---|---|---|---|---|
| | | 1 | 0 | 0 | 1 |

**Pre-requisite: NIL**

**Course Objectives:**

1. To gain a strong foundation in basic mathematical concepts, and become proficient in performing calculations, solving equations, and manipulating numerical data.
2. To apply quantitative techniques in engineering, managerial, and real-life problem contexts. Through practice and exposure to different types of questions, students will learn time-saving techniques and develop strategies to effectively manage time during exams or real-world scenarios.
3. To prepare for various competitive exams, such as aptitude tests for job placements, college admissions tests for higher education (GATE/CAT/GRE/GMAT) and government entrance exams. The course will provide the necessary knowledge and practice to perform well in these exams.

**Course Outcome:** After completion of the course, the student will be able to

1. **understand** fundamental concepts of numbers to solve quantitative aptitude problems.
2. **apply** percentage concepts to accurately calculate and **analyze** profit, loss, discounts, and value changes in real-life and business- Apply concepts of Simple Interest and Compound Interest to analyze and solve quantitative and financial problems effectively in academic, real-life, and competitive exam scenarios.
3. apply the principles of ratio, proportion, and mixtures to **analyze** and solve real-world quantitative problems involving comparisons, allocations, and blending of quantities.
4. apply the principles of Arithmetic's to **analyze** motion related problems, efficiency and work distribution to **evaluate** practical real-life science, engineering and quantitative problems with accuracy.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 1 | - | 1 | - | - | - | - | - | - | 2 |
| CO2 | 3 | 1 | - | 1 | - | - | - | - | - | - | 2 |
| CO3 | 3 | 1 | - | 1 | - | - | - | - | - | - | 2 |
| CO4 | 3 | 1 | - | 1 | - | - | - | - | - | - | 2 |

| Unit 1 | Number Systems-I | 3 hours |
|---|---|---|

Types of numbers: (Natural, Whole, Integers, Rational, Irrational, Real), Divisibility rules, LCM & HCF (Methods and Applications), Unit digit & last digits, concepts, Assumptions and Simplifications.

| | | |
|---|---|---|
| **Unit 2** | **Percentages & Applications** | **3 hours** |
| Basic percentage concepts, Value changes & comparison, Profit, loss & discount, marked price, cost price, selling price relationships, Simple and Compound Interest. | | |
| **Unit 3** | **Ratio & Proportion, Averages & Mixtures** | **3 hours** |
| Ratio & Proportion (Definition of Ratio and Proportion. Type of Proportion (Direct Proportion, Inverse Proportion, Continued Proportion), Definition of Variance (Direct and Inverse), Problem related to age, coins, expenditure and saving, Partnerships). Averages & Mixtures (Weighted averages, Alligation method, Concentration mixing problems) | | |
| **Unit 4** | **Time, Speed and Distance (TSD)** | **3 hours** |
| Basic concepts, definition and terminology related to TSD, Direct and Indirect relation of TSD, Unit Conversions, Average Speed, Relative Speed, Problem on Trains, Problem on Boat & Stream, Race and Games | | |
| **Unit 5** | **Time & Work** | **3 hours** |
| Introduction to the Time and Work, Work and Wages, Problems based on Efficiency, Problems based on Time and work, Problems based on Pipe and Cistern. | | |
| | **Total Lecture hours** | **15 hours** |

**Textbook**

1. Booklet provided by the Aptitude Team, Training Division -CRPC, KIET.

**Reference Books:**

1. "Quantitative Aptitude for Competitive Examinations" by R.S. Aggarwal, S. Chand Publication.
2. "How to Prepare for Quantitative Aptitude for the CAT" by Arun Sharma, TMH Publication.
3. "Magical Book on Quicker Maths" by M. Tyra, Published by BSC Publishing Co. Pvt. Ltd
4. "Quantitative Aptitude Quantum CAT" by Sarvesh K. Verma, Arihant Publication.

**Reference Websites:**

1. https://www.indianbix.com
2. https://www.prepinsta.com
3. https://www.talentsprint.com
4. https://www.careeride.com
5. https://www.smartkeeda.com

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| **MSE** | **CA** | **ESE** | **Total Marks** |
| - | 25 | 25 | 50 |
| 25 | | | |

| **Course Code: HS204L** | **Course Name: Soft Skills for Employability - 1** | **L** | **T** | **P** | **C** |
|---|---|---|---|---|---|
| | | **2** | **0** | **0** | **2** |

**Pre-requisite: Students should have basic knowledge of English grammar and should have cleared the Soft Skills Essentials 1 & 2 of B. Tech 2nd year.**

**Course Objectives:**

To enhance the verbal aptitude of the students by strengthening their proficiency in English grammar, vocabulary, and comprehension skills, enabling them to communicate effectively in academic, professional, and competitive examination contexts.

**Course Outcome:** After completion of the course, the student will be able to:

1. Apply rules of English grammar and sentence structure to construct clear, accurate, and grammatically coherent sentences.
2. Demonstrate enhanced vocabulary usage.

3. Analyze and interpret written passages to answer questions based on reading comprehension, sentence completion, and critical reasoning.
4. Maximize the ability to compose passages solve verbal ability questions commonly asked in placement and recruitment assessments, thereby enhancing overall employability readiness.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | - | - | - | - | - | - | - | 1 | 3 | - | 2 |
| CO2 | - | - | - | - | - | - | - | 1 | 3 | - | 2 |
| CO3 | - | - | - | - | - | - | - | 1 | 3 | - | 2 |
| CO4 | - | - | - | - | - | - | - | 1 | 3 | - | 2 |

| **Unit 1** | **About Grammar Skills** | **8 hours** |
|---|---|---|

Tenses and Concord – Concept Revision followed by Practice Exercises on Moodle, Active and Passive voice, Direct & Indirect Narration – Concept Revision followed by Practice Exercises on Moodle, Determiners & Modifiers – Concept Revision followed by Practice Exercises on Moodle, Articles, Conjunctions, Interjections, and Prepositions – Concept Revision followed by Practice Exercises on Moodle

| **Unit 2** | **Vocab Building** | **7 hours** |
|---|---|---|

Parallelism, Sentence Improvement & Completion, Conditional Sentences (Practice Exercises on Moodle) Vocabulary – Concept of roots & Affixes, synonyms & antonyms, Idioms & Phrases, Analogies, One-word substitutions, odd-man out, spell check, phrasal verbs (practice exercises on Moodle) Vocabulary.com - Learn Words - English Dictionary

| **Unit 3** | **Reading Ability** | **7 hours** |
|---|---|---|

Reading Comprehension & Cloze tests (Practice Exercises on Moodle) https://www.indiabix.com/ Sentence Rearrangement (Jumbled Sentences/Para Jumbles), Missing Sentences (Practice Exercises on Moodle)

| **Unit 4** | **Writing Ability** | **8 hours** |
|---|---|---|

Essentials of Writing: Do's and Don'ts of Writing, Writing a Paragraph and Essay Writing (On Word Document checked by Grammarly and display of some sample paragraphs –Poor and good both), Write & Improve with Cambridge

| | **Total Lecture hours** | **30 hours** |
|---|---|---|

**Textbook**


1. Lewis, Norman. *Word Power Made Easy*. Goyal Publishers, New Delhi.
2. Wren, P. C., & Martin, H. *High School English Grammar and Composition*. S. Chand Publishing, New Delhi.
3. Sharma, Arun, & Upadhyay, Meenakshi. *How to Prepare for Verbal Ability and Reading Comprehension*. McGraw Hill Education, New Delhi.
4. Aggarwal, R. S., & Aggarwal, Vikas. *Objective General English*. S. Chand Publishing, New Delhi.


**References:**


1. https://www.indiabix.com/online-test/verbal-ability-test/
2. https://prepinsta.com/
3. https://www.vocabulary.com/
4. https://www.bbc.co.uk/learningenglish


**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| **MSE** | **CA** | **ESE** | **Total Marks** |
| - | 100 | - | 100 |
| 100 | | | |

| Course Code: IT401B | Course Name: Advance Databases | L | T | P | C |
|---|---|---|---|---|---|
| | | 3 | 0 | 2 | 4 |

**Pre-requisite: Database System**

**Course Objectives:**

The course aims to provide students with advanced knowledge of database programming, modern NoSQL database systems, and graph database technologies used in current industry applications. The course enables students to develop procedural database programs using PL/SQL for enterprise-level transaction processing, design and manage document-oriented databases using MongoDB for flexible and scalable data storage, and model and analyze highly connected data using Neo4j graph databases and Cypher query language. The course focuses on database design, query optimization, data modeling, data processing, and real-world database application development to prepare students for modern data-driven software systems and multi-model data architectures.

**Course Outcome:** After completion of the course, the student will be able to

1. Apply PL/SQL concepts- cursors, stored procedures, packages, and triggers for database applications.
2. Understand the document-oriented database solutions using MongoDB including CRUD operations, schema design, indexing, and query operations.
3. Apply the advanced MongoDB features in modern database applications.
4. Elaborate graph-based data models and interpret relationship queries using Neo4j and Cypher query language for data applications.
5. Use advance database tools for demonstrating mini projects.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 2 | 2 | 2 | - | 1 | 2 | 1 | 1 | 2 |
| CO2 | 3 | 2 | 2 | 1 | 3 | - | 1 | 2 | 1 | - | 2 |
| CO3 | 3 | 2 | 2 | 2 | 3 | - | 1 | 2 | 2 | 1 | 2 |
| CO4 | 3 | 3 | 2 | 2 | 3 | - | 1 | 2 | 2 | 1 | 2 |
| CO5 | 3 | 3 | 3 | 3 | 3 | 1 | 2 | 3 | 2 | 2 | 2 |

**Unit 1 PL/SQL** **15 hours**

Introduction to PL/SQL, PL/SQL architecture, advantages of PL/SQL, structure of PL/SQL blocks, variables and constants, PL/SQL data types, anchored data types (%TYPE, %ROWTYPE), operators and expressions, control structures (IF, CASE), iterative statements (LOOP, WHILE LOOP, FOR LOOP), SQL statements in PL/SQL, SELECT INTO statement, transaction control statements, implicit and explicit cursors, cursor attributes, parameterized cursors, cursor FOR loop, records, exception handling, predefined and user-defined exceptions, RAISE and RAISE_APPLICATION_ERROR, stored procedures, functions, parameter modes, packages, package specification and package body, database triggers, types of triggers, trigger events, and trigger restrictions.

**Mini Projects:**

1. Banking Transaction Management System: To implement database-driven banking operations using PL/SQL procedures, functions, and triggers.
2. Student Result Processing and Grade Automation System: To automate student result calculation and grading using PL/SQL.

**Unit 2 MongoDB Fundamentals and Data Modeling** **20 hours**

Introduction to NoSQL databases, overview of MongoDB architecture, installation and MongoDB environment, BSON document structure, collections and documents, CRUD operations (insert, update, delete, find), query operators, projection, sorting and limiting results, indexing concepts and index types, schema design principles, embedding vs referencing documents, data modeling patterns, working with arrays and nested documents, cursor handling, and basic database administration operations.

**Mini Projects:**

1. E-Commerce Product Catalog Management: To design and implement a document-based database using MongoDB focusing on CRUD operations, schema design, and indexing.

2. Library Management Document Database: To design a document-oriented database focusing on schema design, embedded documents, CRUD operations, and indexing.

| Unit 3 | Advanced MongoDB Operations and Performance | 20 hours |
|---|---|---|

Aggregation framework and aggregation pipeline stages, advanced querying techniques, text search basics, replication concepts and replica sets, sharding fundamentals, transactions in MongoDB, performance optimization and query profiling, backup and restore concepts, security basics including authentication and authorization, monitoring tools overview, deployment considerations, and introduction to MongoDB integration with applications and modern data-driven systems.

**Mini Projects:**

1. IoT Sensor Data Analytics System: To implement aggregation pipelines, performance optimization concepts, and large data handling using MongoDB.
2. Social Media Analytics and User Activity System: To implement advanced MongoDB features such as aggregation, replication concepts, transactions, and performance optimization.

| Unit 4 | Neo4j Graph Database | 20 hours |
|---|---|---|

Introduction to graph databases and NoSQL graph model, comparison of relational databases and graph databases, property graph model concepts, nodes, relationships, properties, labels, graph database architecture and Neo4j ecosystem overview, installation and Neo4j tools overview, Cypher query language fundamentals, creating and managing nodes and relationships, MATCH, CREATE, MERGE, DELETE and RETURN clauses, filtering, sorting and aggregation in Cypher, path traversal and pattern matching, graph data modeling techniques, schema design in graph databases, indexing and constraints in Neo4j, graph query optimization basics, data import and export concepts, graph analytics basics, shortest path concept, recommendation system modeling, security and access control basics, backup and recovery concepts, performance and scalability concepts, and real-world applications of graph databases.

**Mini Projects:**

1. Social Network Relationship Graph: To design and query relationship-based data using Neo4j graph modeling and Cypher queries.
2. Movie Recommendation Graph System: To implement graph analytics and recommendation logic using Neo4j relationships and path traversal.

| | Total Lecture Hours | 75 hours |
|---|---|---|

**Textbook:**

1. PL/SQL by Example — Benjamin Rosenzweig, Elena Silvestrova (Oracle-based PL/SQL programming reference)
2. MongoDB: The Definitive Guide — Kristina Chodorow, Michael Dirolf
3. Graph Databases — Ian Robinson, Jim Webber, Emil

**Reference Books:**

1. Learning Neo4j — Rik Van Bruggen
2. MongoDB in Action — Kyle Banker

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| MSE | CA | ESE | Total Marks |
| 80 | 20 | 100 | 200 |
| 100 | | | |

| Course Code: AI401B | Course Name: NLP and Large Language Models | L | T | P | C |
|---|---|---|---|---|---|
| | | 2 | 0 | 2 | 3 |

**Pre-requisite: Introduction to AI, Python, Machine Learning Essentials, Deep Learning Essentials**

**Course objective:**

To provide students with the knowledge and skills to understand, develop, and apply modern NLP, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and multimodal Generative AI techniques for real-world applications.

**Course outcomes:**

**CO1:** Explain NLP fundamentals including embeddings and language modeling.
**CO2:** Implement transformer-based models for NLP tasks.
**CO3:** Analyze architecture and training of Large Language Models.
**CO4:** Design RAG and LLM-based application systems.
**CO5:** Develop multimodal and generative AI systems.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO-PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 1 | 1 | 1 | 2 | 1 | 1 | 1 | 1 | 1 | 1 |
| CO2 | 2 | 2 | 2 | 1 | 2 | 1 | 1 | 1 | 2 | 2 | 1 |
| CO3 | 2 | 2 | 1 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | 1 |
| CO4 | 2 | 2 | 1 | 2 | 2 | 1 | 1 | 1 | 1 | 1 | 1 |
| CO5 | 2 | 1 | 2 | 1 | 2 | 2 | 2 | 2 | 3 | 3 | 2 |

| Unit1 | Foundations of NLP for Generative AI | 12 hours |
|---|---|---|

Text Processing Pipeline, Tokenization (WordPiece, BPE, SentencePiece), Word Embeddings (Word2Vec, GloVe, FastText), Contextual Embeddings (ELMo, BERT intro), Language Modeling (n-gram → Neural LM), Evaluation Metrics (Perplexity, BLEU, ROUGE)

**Case Studies:**

- Language Modeling on Shakespeare Dataset

**Practical:**

- Compare static vs contextual embeddings

| Unit 2 | Transformers & Modern NLP | 12 hours |
|---|---|---|

Self-Attention (Mathematical Derivation), Multi-Head Attention, Positional Encoding, Encoder-Decoder Architecture, Pretraining Objectives (MLM, CLM), Fine-tuning Strategies, HuggingFace Ecosystem

**Case Studies:**

- Sentiment Analysis using BERT

**Practical:**

- Build Text Classifier with DistilBERT

| Unit 3 | Large Language Models (LLMs) | 12 hours |
|---|---|---|

Evolution: GPT-1 → GPT-4, Scaling Laws, Instruction Tuning, RLHF (Reinforcement Learning from Human Feedback), Prompt Engineering (Zero-shot, Few-shot, CoT), LLM Evaluation (Hallucination, Bias, Safety)

**Case Studies:**

- Prompt Engineering Experiments
- Instruction Fine-Tuning Demo

**Practical:**

- Build Chatbot using OpenAI API

Evaluate LLM outputs

| Unit 4 | Retrieval-Augmented Generation (RAG) & LLM Systems | 12 hours |
|---|---|---|

RAG Architecture, Vector Databases (FAISS, Pinecone, ChromaDB), Embedding Models, Document Chunking & Indexing, LangChain & LlamaIndex, Agents & Tool Calling, Evaluation of RAG Systems

**Case Studies:**

- Enterprise Knowledge Assistant

**Practical:**

- Deploy FastAPI-based LLM App
- Implement Vector Search

| Unit 5 | Advanced Generative AI & Multimodal Systems | 12 hours |
|---|---|---|

Generative Models Overview, Autoencoders & VAEs, GANs, Diffusion Models (DDPM), Text-to-Image (Stable Diffusion overview), Multimodal Models (CLIP, Vision-Language Models), AI Safety & Alignment, Responsible AI & Ethics, Deployment & MLOps for GenAI

**Case Studies:**

- Diffusion Model Demo

**Practical:**

- Build Image Caption Generator

| | |
|---|---|
| **Total Lecture Hours** | **60** |

**Textbook:**

1. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, **Deep Learning**, MIT Press, 2016.
2. Daniel Jurafsky and James H. Martin, **Speech and Language Processing** (3rd Edition, draft), Pearson. Available: Speech and Language Processing Book
3. **Stanford CS224n: Natural Language Processing with Deep Learning**, Stanford University. Course materials available at: Stanford CS224n
4. **Stanford CS324: Large Language Models**, Stanford University. Course materials available at: Stanford CS324

**Reference Books:**


1. Hugging Face. (n.d.). *Hugging Face Documentation*. Available: https://huggingface.co/docs
2. OpenAI. *GPT-4 Technical Report*. 2023. Available: https://arxiv.org/abs/2303.08774
3. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). *Attention Is All You Need*. *Advances in Neural Information Processing Systems, 30*. https://arxiv.org/abs/1706.03762


**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| **MSE** | **CA** | **ESE** | **Total Marks** |
| **60** | **15** | **75** | **150** |
| **100** | | | |

| **Course Code: AI316B** | **Course Name: Software System Engineering and DevOps** | **L** | **T** | **P** | **C** |
|---|---|---|---|---|---|
| | | **2** | **0** | **2** | **3** |

**Pre-requisite:** Programming Fundamentals, OOP, Database Basics, Operating Systems, Computer Networks, and Basic Web Technologies

**Course Objectives:**

To enable students to apply software engineering and DevOps practices for building, deploying, automating, and managing scalable cloud-native applications using modern tools and technologies.

**Course outcomes:**

CO1: Introduce the fundamental principles of Software Engineering, including requirements engineering, system design, modeling techniques, testing strategies, maintenance processes, and Agile methodologies for structured software development.
CO2: Develop the ability to implement DevOps practices and Linux fundamentals, including shell scripting, user and file management, and version control using Git for collaborative and automated software workflows
CO3: Enable students to implement containerization and cloud-based deployments, using Docker, cloud computing constructs, and AWS services to build scalable and portable applications.
CO4: Familiarize students with container orchestration and CI/CD automation, including Kubernetes, Docker Swarm, Helm, Ansible, and pipeline integration for continuous integration and continuous delivery.
CO5: Expose learners to Infrastructure as Code (IaC), monitoring, and advanced DevOps practices, including Terraform, observability tools, serverless computing, and cloud-native architectural best practices.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **CO1** | 3 | 2 | 3 | 2 | 2 | 1 | 1 | 1 | 1 | 1 | 2 |
| **CO2** | 2 | 2 | 3 | 2 | 2 | 1 | 1 | 1 | 2 | 1 | 2 |
| **CO3** | 2 | 3 | 3 | 3 | 3 | 2 | 1 | 1 | 2 | 1 | 2 |
| **CO4** | 2 | 3 | 3 | 3 | 3 | 2 | 1 | 1 | 3 | 2 | 3 |
| **CO5** | 2 | 3 | 3 | 3 | 3 | 2 | 2 | 2 | 2 | 2 | 3 |

| **Unit 1** | **Software Engineering Fundamentals** | **12 hours** |
|---|---|---|

Introduction to Software Engineering, Requirements Engineering, Design & Modeling, Testing & Maintenance, Introduction to Agile

**Case Study:** Development of an Online Food Delivery Application

**Background**
A startup company wants to develop an Online Food Delivery Application similar to Swiggy or Zomato. The application should allow customers to browse restaurants, place orders, track delivery, and make payments online.

| **Unit 2** | **Fundamentals of DevOps and Linux** | **12 hours** |
|---|---|---|

Introduction to DevOps, Operating Systems and VM Basics, Linux Fundamentals and CLI Commands, User and File Management, Shell Scripting and Bash, Version Control with Git
**Case Study:** Managing a Web Server Using Linux and DevOps Practices
**Background**
A company hosts a company website on a Linux server. The DevOps team is responsible for managing the server and ensuring smooth operations.

| **Unit 3** | **Containerization and Cloud Fundamentals** | **12 hours** |
|---|---|---|

Introduction to Containerization, Docker Fundamentals, Docker Deep Dive, Cloud Computing Fundamentals, AWS/Cloud Constructs, Introduction to ECS
**Case Study:** Deploying an E-commerce Application Using Docker and AWS
**Background**
An e-commerce company wants to deploy its application efficiently and ensure it runs consistently across environments.

| **Unit 4** | **Container Orchestration and CI/CD** | **12 hours** |
|---|---|---|

Container Orchestration Concepts, Docker Swarm, Kubernetes Fundamentals, Package Management with Helm, Configuration Management with Ansible, CI/CD Fundamentals
**Case Study:** Automating Deployment of a Banking Application
**Background**
A bank develops a digital banking platform and wants automated deployment and scalability

| **Unit 5** | **Infrastructure as Code, Monitoring, and Advanced Topics** | **12 hours** |
|---|---|---|

Infrastructure as Code (IaC), Terraform, Monitoring and Observability, Serverless Computing Deep Dive, Advanced AWS Services, DevOps Best Practices and Future Trends
**Case Study:** Automated Cloud Infrastructure for a Streaming Platform
**Background**
A video streaming platform needs scalable infrastructure to support millions of users.

| | |
|---|---|
| **Total Lecture Hours** | **60 hours** |

**Textbook:**
1. Saito, H., & Rushgrove, G. Continuous Delivery with Docker and Kubernetes: Delivering Containerized Applications in CI/CD Pipelines. Packt Publishing, 2021.
2. D'Anna, D. (2020). Kubernetes and Docker – An Enterprise Guide: Effectively Containerize Applications, Integrate Enterprise Systems, and Scale Applications in Your Enterprise. Packt Publishing.

**Reference Books:**
1.The DevOps Handbook – Gene Kim, Patrick Debois, John Willis, Jez Humble

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| **MSE** | **CA** | **ESE** | **Total Marks** |
| **60** | **15** | **75** | **150** |
| **75** | | | |

# Professional Elective-III Courses Detail Syllabus – V Sem

| Course Code: AI405E | Course Name: Agentic AI | L | T | P | C |
|---|---|---|---|---|---|
| | | 3 | 0 | 2 | 4 |

**Pre-requisite:** Introduction to AI, Python, Machine Learning Essentials, Deep Learning Essentials, Computer Vision, GenAI

**Course Objective:** To provide students with the fundamental concepts and practical techniques required to design, implement, and evaluate autonomous Agentic AI systems incorporating reasoning, memory, multi-agent collaboration, and ethical deployment.

**Course outcomes:**

CO1: Explain the fundamentals of Agentic AI and design appropriate intelligent agent architectures.
CO2: Implement reasoning-based agents using planning, tool calling, and task decomposition techniques.
CO3: Develop memory-enabled agents using vector databases, RAG, and knowledge integration methods.
CO4: Design and coordinate multi-agent systems for collaborative task execution.
CO5: Deploy and evaluate Agentic AI systems with safety, ethics, monitoring, and human oversight mechanisms.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 1 | 1 | 1 | 2 | 1 | 1 | 1 | 1 | 1 | 1 |
| CO2 | 2 | 2 | 2 | 1 | 2 | 1 | 1 | 1 | 2 | 2 | 1 |
| CO3 | 2 | 2 | 1 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | 1 |
| CO4 | 2 | 2 | 2 | 2 | 2 | 1 | 1 | 1 | 1 | 1 | 1 |
| CO5 | 2 | 1 | 1 | 1 | 2 | 2 | 2 | 2 | 3 | 3 | 2 |

**Unit 1 Foundations of Agentic AI** — **15 hours**

Evolution from Reactive AI → Generative AI → Agentic AI, Intelligent Agents: Definition and Properties, PEAS Framework (Performance, Environment, Actuators, Sensors), Types of Agents: Simple Reflex Agents, Model-Based Agents, Goal-Based Agents, Utility-Based Agents, Learning Agents, LLMs as Agents, Agent Architecture Overview.

Case Studies 1: Autonomous Customer Support Agent

- Design an LLM-powered agent that:
- Understands user query
- Retrieves company policy from documents
- Responds autonomously
- Compare simple chatbot vs agent with reasoning loop

**Unit 2 Reasoning, Planning & Tool Use** — **15 hours**

Chain-of-Thought Reasoning, ReAct Framework (Reason + Act), Tool Calling & Function Calling, Planning Algorithms in Agents, Task Decomposition, Reflection & Self-Correction.

Case Study 2: AI Research Assistant Agent

- Searches research papers
- Summarizes key findings
- Extracts citations
- Creates literature review draft

**Unit 3 Memory & Retrieval-Augmented Agents** — **15 hours**

Types of Memory: Short-term memory, Long-term memory, Vector Databases (FAISS, Pinecone, Chroma), Retrieval-Augmented Generation (RAG), Context Management, Episodic vs Semantic Memory, Knowledge Graph Integration.

Case Study 3: AI Academic Advisor Agent using Lightning AI and neo4j

- Maintains student history
- Suggests courses based on past performance
- Uses RAG for syllabus retrieval
- Stores and updates long-term student profile

| Unit 4 | Multi-Agent Systems & Collaboration | 15 hours |
|---|---|---|

Multi-Agent Architecture, Role-Based Agents, Communication Protocols, Coordination & Conflict Resolution, Swarm Intelligence Concepts, AutoGen & CrewAI Frameworks

Case Study 4: Startup Simulation – Multi-Agent Business System
- CEO Agent (Strategy)
- Marketing Agent
- Finance Agent
- Product Agent

| Unit 5 | Deployment, Ethics & Real-World Applications | 15 hours |
|---|---|---|

Agent Safety & Guardrails, AI Alignment, Monitoring & Logging, Human-in-the-Loop Systems, Responsible AI, Deployment using: Streamlit, FastAPI, Cloud platforms, Evaluation Metrics for Agents, Limitations & Risks of Agentic AI.

Case Study 5: Autonomous Hiring Agent
- Screens resume
- Evaluates skills
- Schedules interviews
- Maintains fairness and bias checks
- Includes human approval before final decision

| Total Lecture Hours | 75 hours |
|---|---|

**Textbook:**

1. **Building Agentic AI Systems, O'Reilly-2025**
2. https://www.ibm.com/training/course/introduction-to-agentic-ai-CLPCC202501

**Reference Books:**

1. HuggingFace Documentation
2. https://www.databricks.com/training/catalog/ai-agent-fundamentals-4482
3. https://learn.deeplearning.ai/courses/agentic-ai/information

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| MSE | CA | ESE | Total Marks |
| 80 | 20 | 100 | 200 |
| 100 | | | |

| Course Code: AI406E | Course Name: Cloud Security Foundations | L | T | P | C |
|---|---|---|---|---|---|
| | | 3 | 0 | 2 | 4 |

**Pre-requisite: Fundamentals of Cyber Security**

**Course Objectives:** To provide students with the knowledge and skills to analyze and implement secure cloud computing environments using cloud security principles, AWS security services, and governance best practices.

**Course outcomes:**

**CO1:** Analyze cloud-specific threats, shared responsibility models, and governance requirements to assess security risks in cloud environments.

**CO2:** Analyze and compare security architectures, identity models, and control mechanisms across Amazon Web Services, Microsoft Azure, and Google Cloud Platform.

**CO3:** Analyze and configure identity, access control, and network security mechanisms in AWS to secure cloud infrastructure.

**CO4:** Analyze and implement data protection, encryption, logging, and monitoring mechanisms in AWS environments.

**CO5:** Analyze security events and apply appropriate incident response and governance mechanisms within AWS environments.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 1 | 1 | 2 | 2 | 2 | – | – | – | 1 |
| CO2 | 3 | 3 | 2 | 1 | 2 | 1 | 1 | – | – | – | 1 |
| CO3 | 3 | 3 | 3 | 2 | 3 | 1 | 2 | 1 | – | – | 1 |
| CO4 | 3 | 3 | 3 | 2 | 3 | 2 | 2 | – | 1 | – | 1 |
| CO5 | 3 | 3 | 2 | 3 | 3 | 2 | 3 | 1 | 2 | 1 | 2 |

| Unit 1 | Cloud Security Introduction | 15 hours |
|---|---|---|

**Theory:**
Cloud threat landscape and cloud-native attack vectors; cloud attack surface analysis; identity-based attacks and credential compromise; cloud misconfigurations and data exposure; insecure APIs and application-layer risks; insider threats and supply chain risks in cloud; security design principles including least privilege, defense in depth, zero trust architecture, segregation of duties, and secure configuration baselines; shared responsibility model from a security accountability perspective across IaaS, PaaS, and SaaS; cloud security governance concepts; risk management in cloud environments; compliance frameworks overview (ISO 27001, SOC 2, CIS Benchmarks); introduction to cloud security posture management.
**Lab 1:** Vulnerability identification and cloud risk mapping using OWASP Juice Shop.

| Unit 2 | Security Across Major Cloud Platforms | 15 hours |
|---|---|---|

**Theory:**
Comparative security architecture of major cloud providers; identity and access management models across Amazon Web Services, Microsoft Azure, and Google Cloud Platform; RBAC and policy structures across platforms; multi-account and multi-project security organization; network isolation models (VPC, VNet, global VPC concepts); native logging and monitoring services comparison; threat detection and security center services comparison; encryption and key management services overview across providers; governance and policy enforcement mechanisms; shared responsibility comparison across providers; overview of security capabilities in IBM Cloud, Oracle Cloud Infrastructure, and Alibaba Cloud; multi-cloud security challenges and interoperability risks.
**Lab 2:** Web application security scanning and analysis using OWASP ZAP with cloud control mapping exercise.

| Unit 3 | AWS Identity and Infrastructure Security | 15 hours |
|---|---|---|

**Theory:**
Deep dive into AWS shared responsibility model; IAM architecture; users, groups, roles, and policies; policy structure and permission evaluation logic; managed policies and custom policies; resource-based policies; cross-account access and role assumption; multi-account strategy using AWS Organizations; VPC security architecture; subnet segmentation; security groups and network ACLs; internet gateway and NAT gateway security considerations; bastion host architecture; load balancer security basics; infrastructure hardening practices in AWS.
**Lab 3:** Using Resource-Based Policies to Secure an S3 Bucket.
**Lab 4:** Securing VPC Resources by Using Security Groups.

| Unit 4 | AWS Data Security and Monitoring | 15 hours |
|---|---|---|

**Theory:**
Data classification and protection strategies in AWS; data at rest and data in transit protection mechanisms; S3 security best practices; S3 bucket policies and block public access controls; encryption using AWS managed keys and customer managed keys; introduction to AWS Key Management Service (KMS); key policies and key rotation concepts; secure storage practices for EBS and RDS; secrets management overview; logging fundamentals; AWS CloudTrail architecture; Amazon CloudWatch logging and monitoring; introduction to AWS Security Hub.
**Lab 5:** Protecting Data (S3 and Encryption).
**Lab 6:** Configure Monitoring and Logging in AWS Environment.

| Unit 5 | AWS Threat Detection, Incident Response, and Governance | 15 hours |
|---|---|---|

**Theory:**
Cloud incident response lifecycle; incident detection and containment in AWS; AWS Config and configuration compliance concepts; introduction to GuardDuty, Macie, Detective, Inspector; centralized security monitoring concepts; remediation strategies; forensic considerations in cloud environments; governance using AWS Organizations; service control policies (conceptual); compliance and auditing in AWS; secure architecture integration and review; best practices for continuous security improvement in AWS environments.

**Lab 7:** Remediating an Incident Using AWS Config.
**Lab 8:** Capstone Secure AWS Architecture Implementation.

| | |
|---|---|
| **Total Lecture Hours** | **75 hours** |

**Textbook and References:**

1. Dotson, C. (2023). Practical Cloud Security: A Guide for Secure Design and Deployment. O'Reilly Media.
2. Scott, S. (2023). AWS Certified Security – Specialty (SCS-C02) Exam Guide (2nd ed.). Packt Publishing.

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| **MSE** | **CA** | **ESE** | **Total Marks** |
| 80 | 20 | **100** | **200** |
| **100** | | | |

| **Course Code: AI407E** | **Course Name: Advanced Analytics and Visualization** | **L** | **T** | **P** | **C** |
|---|---|---|---|---|---|
| | | **3** | **0** | **2** | **4** |

**Pre-requisite:** Data Engineering Fundamentals

**Course Objectives:** To provide students with the knowledge and skills to prepare, visualize, publish, and manage data using Tableau while following industry best practices for analytics and governance.

**Course Outcomes (COs): At the end of the course, student will be able to:**

CO1: Explain Tableau platform, certification tracks, and exam domains.
CO2: Perform data connection, cleaning, and transformation using Tableau Desktop & Tableau Prep.
CO3: Build dashboards and publish/manage content on Tableau Server/Cloud.
CO4: Apply operational concepts such as extract refresh, scheduling, and monitoring.
CO5: Implement governance, permissions, roles, and best practices.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **CO1** | 2 | 1 | 1 | - | 1 | - | 1 | - | 1 | - | 2 | 2 |
| **CO2** | 2 | 3 | 3 | 2 | 2 | 1 | 1 | 2 | 1 | 1 | 1 | 3 |
| **CO3** | 2 | 3 | 3 | 3 | 2 | 1 | 2 | 2 | 1 | 1 | 1 | 3 |
| **CO4** | 2 | 3 | 3 | 3 | 3 | 1 | 1 | 1 | 2 | 2 | 1 | 3 |
| **CO5** | 2 | 3 | 2 | 3 | 2 | 2 | 3 | 1 | 2 | 1 | 1 | 3 |

| **Unit 1** | **Connecting, Preparing & Transforming Data** | **15 hours** |
|---|---|---|

Connecting to data sources: files, databases, cloud platforms
Live vs Extract connections
Data cleaning: pivot, split, rename fields, change data types
Joins, relationships, unions
Data quality checks: completeness, consistency, accuracy
Tableau Prep Builder: flows, aggregation, filtering, joins, publishing outputs
Hands-on:
Build a Tableau Prep flow and export a .hyper file
Connect multiple data sources in Desktop
Create data source filters and extract filters
Case Study:
Retail dataset: end-to-end Prep workflow.

| **Unit 2** | **Exploring, Analyzing & Visualizing Data** | **15 hours** |
|---|---|---|

Calculated fields: numeric, string, date, logical
LOD expressions: FIXED, INCLUDE, EXCLUDE
Table calculations: running total, moving average, percent-of-total, ranking

Visualizations: bar, line, scatter, maps, density maps, dual-axis charts
Analytics tools: reference lines, trend lines, forecasting
Filters: dimension, measure, context, Top-N, wildcard
Hands-on:
Build KPI dashboard using table calculations
Create parameter-driven visualizations
Build geospatial maps

| Unit 3 | Dashboards, Stories, Publishing & Content Management | 15 hours |
|---|---|---|

Dashboard design: containers, layout, device optimization
Dashboard actions: filter, highlight, navigation
Story building: multi-page guided narratives
Publishing to Tableau Server/Cloud
Managing content: scheduling refreshes, subscriptions, alerts
Hands-on:
Build and publish dashboards
Configure refresh schedules and alerts
Case Study:
Build and publish a Marketing Performance Dashboard with scheduled refresh.

| Unit 4 | Server Administration, Permissions & Governance | 15 hours |
|---|---|---|

Tableau Server architecture: processes, topology
Identity store, user roles, licensing
Monitoring, job management, performance tuning
Permissions: users, groups, projects, sites
Governance: data lineage, content certification, row-level security
Hands-on:
Create users & groups
Assign permissions
Manage extract refresh schedules

| UNIT 5 | Consultant/Architect Practices + Certification Preparation | 15 hours |
|---|---|---|

Assessment of organizational requirements
Planning architecture and data connection strategies
Troubleshooting dashboards and performance issues
Enterprise governance and deployment patterns
Complete review of Tableau Data Analyst exam blueprint
Exam registration, structure, and preparation strategies
Mini-Project:
End-to-end BI pipeline:
Prep Flow → Desktop Visualizations → Dashboard → Publish → Permissions → Scheduling → Monitoring.

| | |
|---|---|
| **Total Lecture Hours** | **75** |

**Textbooks and References:**

1. Milligan, J. N. (2025). Learning Tableau 2025: Leverage Tableau's Newest Features to Revolutionize Your Data Storytelling with AI-Enhanced Insights (6th ed.). Packt Publishing. ISBN: 978-1835886786
2. Cooney, H., & Jones, D. (2024). Tableau Certified Data Analyst Certification Guide: Ace the Tableau Data Analyst Certification Exam with Expert Guidance and Practice Material (1st ed.). Packt Publishing. ISBN 978-1803243467
3.Tableau Software. Tableau Help. Salesforce, Inc., https://help.tableau.com/

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| **MSE** | **CA** | **ESE** | **Total Marks** |
| **80** | **20** | **100** | **200** |
| **100** | | | |

# Practical Courses Detail Syllabus – V Sem

| Course Code: IT302P | Course Name: Computer Networks Lab | L | T | P | C |
|---|---|---|---|---|---|
| | | 0 | 0 | 2 | 1 |

**Pre-requisite: Computer Fundamentals and C programming**

**Course Objectives:**

The objective of this course is to provide students with a theoretical and practical base in computer networks issues and prepare the students for easy transfer from academia into practical life.

**Course Outcome:** After completion of the course, the student will be able to

1. Understand the fundamental concepts of computer networking and Network topologies.
2. Analyze different types of network devices and simple computer networks.
3. Implement the basic network commands and use techniques, skills, and modern networking tools necessary for engineering practice.
4. Analyze the impact of computer network technologies on society and professional development.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 2 | - | - | 2 | - | - | - | - | 1 | 2 |
| CO2 | 3 | 2 | - | - | 2 | - | - | - | - | 1 | 2 |
| CO3 | 3 | 3 | 2 | 2 | 2 | - | - | - | 2 | 1 | 2 |
| CO4 | - | 2 | - | - | - | 3 | - | - | - | - | 3 |

**List Of Practical's (Indicative & Not Limited To)**

1. To study of LAN using different Network topologies and different network connecting devices like Hub, Switch, Router etc.
2. To study and learn the handling of networking hardware like: NIC card, RJ-45 connector, crimping tool, CAT-6 cable and Do the following Cabling works in a network (T568A and T568B) a) Cable Crimping b) Standard Cabling and c) Cross Cabling d) IO connector crimping e) Testing the crimped cable using a cable tester.
3. Running and using services/commands like ifconfig, ping, traceroute, nslookup, telnet, ftp, etc.
4. Establish a physical peer to peer network connection using two or more systems using network interface card, network cables, Switch and Router in a LAN.
5. To install the network packet analysis tool Wireshark and also capture and analyze the local traffic.
6. Simulate and Configure a Network topology using packet tracer software.
7. Implement the data link layer framing methods such as character count, character stuffing and bit stuffing.
8. Implement the error correcting code Cyclic Redundancy Check (CRC) of data link layer using various polynomials like CRC-CRC 12, CRC 16 and CRC CCIPP.
9. Implement a program to determine if the IP address is in Class A, B, C, D, or E and translate dotted decimal IP address into 32-bit address.
10. Implement Dijkstra algorithm to compute the shortest path through a given graph.

**Total Lecture Hours: 30 hrs.**

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| MSE | CA | ESE | Total Marks |
| - | 25 | 25 | 50 |
| 25 | | | |

| Course Code: CS313P | Course Name: Problem Solving Lab-1 | L | T | P | C |
|---|---|---|---|---|---|
| | | 0 | 0 | 4 | 2 |

**Pre-requisite: NIL**

**Course Objectives:**

1. To familiarize students with the basic and advance of Problem Solving.
2. To learn modern techniques to solve DSA Problems.

**Course Outcome:** After completion of the lab, the student will be able to

1.Apply fundamental programming and mathematical reasoning
2.Implement fundamental problem-solving techniques
3. Analyze and optimize algorithmic solutions
4. Integrate problem-solving techniques for programming applications

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 1 | – | 1 | – | – | – | – | – | 1 |
| CO2 | 3 | 3 | 2 | – | 2 | – | – | – | – | – | 1 |
| CO3 | 2 | 3 | 2 | 1 | 1 | – | – | – | – | – | 1 |
| CO4 | 2 | 2 | 3 | 1 | 3 | – | – | 2 | 1 | 1 | 2 |

**List of Experiments:**

1. Basic Logic Building - 1. Print all divisors 2. Prime check 3. GCD using Euclid
2. Number Theory Basics - 1. Sieve of Eratosthenes 2. Count primes in range 3. Modular arithmetic problems
3. Recursion-I - 1. Factorial/Power recursion 2. Print subsets 3. Fibonacci recursion
4. Recursion-II/Backtracking Basics - 1. N-Queen (4×4) 2. Rat in a maze 3. Print permutations
5. Two Pointer Technique–I - 1. Pair sum 2. Remove duplicates from sorted array 3. Move zeros to end
6. Two Pointer Technique–II - 1. 3-sum 2. Container with most water 3. String palindrome check
7. Sliding Window–I - 1. Max sum subarray (fixed k) 2. Count anagram occurrences 3. Longest substring without repeat
8. Sliding Window–II - 1. Minimum window substring 2. Longest repeating character replacement 3. Fruit into baskets
9. Binary Search–I - 1. First/last occurrence 2. Square root using BS 3. Search in rotated sorted array
10. Binary Search on Answer - 1. Allocate books 2. Aggressive cows 3. Koko eating bananas
11. Greedy–I - 1. Activity selection 2. Minimum coins 3. Jump game
12. Greedy-II - 1. Job sequencing 2. Gas station problem 3. Minimum platforms
13. Bit Manipulation–I - 1. Count set bits 2. Find unique element using XOR 3. Check power of 2
14. Bit Manipulation–II - 1. Subset generation using bits 2. Swap bits in number 3. Count bits from 1 to N

Lab-1 Mini Project - 1. Pattern-based problem sheet (mix) 2. Contest simulation 3. 10 mixed technique problems

| Total Laboratory Hours | 60 hours |
|---|---|

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| MSE | CA | ESE | Total Marks |
| - | 100 | - | 100 |
| 100 | | | |

# Theory Courses Detail Syllabus – VI Sem

| Course Code: AI402L | Course Name: Automata Theory and compiler Design | L | T | P | C |
|---|---|---|---|---|---|
| | | 3 | 0 | 0 | 3 |

**Pre-requisite:** Discrete structures

**Course Objectives:** To enable students to understand and apply compiler design principles for developing lexical analyzers, parsers, code generators, and optimization techniques.

**Course outcomes**:
CO1: Construct and analyze finite automata and regular expressions for lexical pattern recognition in compiler design.
CO2: Design context-free grammars and pushdown automata, and apply LL/LR parsing techniques for syntax analysis.
CO3: Develop syntax-directed definitions, translation schemes, and generate intermediate code for semantic analysis.
CO4: Implement run-time storage management, symbol tables, and generate target machine code with register allocation.
CO5: Apply Turing machine concepts and undecidability theory, and perform code optimization using data-flow analysis.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 3 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 2 |
| CO2 | 3 | 3 | 3 | 2 | 3 | 2 | 3 | 3 | 3 | 3 | 2 |
| CO3 | 3 | 3 | 3 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| CO4 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 2 |
| CO5 | 3 | 3 | 3 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 2 |

**Unit 1 Formal Automata and Compiler Introduction** — **12 hours**

Chomsky Hierarchy,
Alphabets, Strings, Languages,
Finite Automata: DFA, NFA, Epsilon-NFA and equivalence,
DFA Minimization.
Regular Expressions and Regular Languages.
Conversion: RE to FA, FA to RE.
Phases of Compiler
Lexical Analysis: Tokens, Patterns, Input Buffering, Lex/Flex.

**Unit 2 Context-Free Grammars & Compiler Parsing** — **12 hours**

Context-Free Grammars: Derivation, Parse Trees, Ambiguity.
CFG Simplification: ε-productions, Unit productions, Useless symbols.
Left recursion elimination, Left factoring.
Pushdown Automata: DPDA, PDA-CFG equivalence
. Parsing: LL(1) Parsing, Predictive Parsing,
Recursive Descent.
Bottom-Up Parsing: Shift-Reduce, LR(0), SLR(1), CLR(1), LALR(1).

**Unit 3 Syntax Translation in Compiler** — **7 hours**

Syntax-Directed Definitions: S-attributed and L-attributed.
Syntax-Directed Translation Schemes.
Construction of Syntax Trees. Semantic Analysis: Type checking,
Type conversions. Intermediate Code:
Three Address Code, Quadruples, Triples

**Unit 4 Compiler Runtime Code Generation** — **7 hours**

Runtime Storage Management: Stack, Heap,
Activation Records. Parameter Passing Methods.
Symbol Table Management.
Basic Blocks, Flow Graphs.
Code Generation Issues. Register Allocation and Assignment.

| Unit 5 | Computability and Compiler Optimization | 7 hours |
|---|---|---|

Turing Machines: Definitions,
Models, Multi-tape TM, NTM,
Undecidability: Recursive, RE languages,
Halting Problem, Rice's theorem.
Code Optimization: Peephole,
Local Optimization.
Data Flow Analysis: Constant Propagation, Liveness Analysis.

| Total Lecture Hours | 45 hours |
|---|---|

**Textbooks:**

Hopcroft, J. E., Motwani, R., & Ullman, J. D. *Introduction to Automata Theory, Languages, and Computation*. 3rd Edition, Pearson Education, 2008.
Mishra, K. L. P., & Chandrasekharan, N. *Theory of Computer Science: Automata, Languages and Computation*. 4th Edition, PHI Learning Pvt. Ltd., 2022.
Aho, A. V., Lam, M. S., Sethi, R., & Ullman, J. D. *Compilers: Principles, Techniques, and Tools*. 2nd Edition, Pearson Education, 2007.

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| MSE | CA | ESE | Total Marks |
| 60 | 15 | 75 | 150 |
| 75 | | | |

| Course Code: HS205L | Course Name: Aptitude - 4 | L | T | P | C |
|---|---|---|---|---|---|
| | | 1 | 0 | 0 | 1 |

**Pre-requisite: NIL**

**Course Objectives:**

1. To gain a strong foundation in basic mathematical concepts, and become proficient in performing calculations, solving equations, and manipulating numerical data.
2. To apply quantitative methods effectively in interpreting data, optimizing solutions, and making informed decisions in academic, professional, and real-world engineering contexts.
3. To build a strong foundation in basic mathematical concepts including arithmetic, algebra, geometry, mensuration, trigonometry, and data interpretation required for engineering and analytical problem-solving.
4. To prepare for various competitive exams, such as aptitude tests for job placements, college admissions tests for higher education (GATE/CAT/GRE/GMAT) and government entrance exams. The course will provide the necessary knowledge and practice to perform well in these exams.

**Course Outcome:** After completion of the course, the student will be able to

1. Gain a strong foundation in mathematical concepts and demonstrate proficiency in calculations, equation solving, and numerical data manipulation, with the ability to apply these skills to solve mathematical and engineering problems.
2. Apply the principles of permutation, combination, and probability to analyze uncertainty and solve counting and chance-based problems relevant to mathematical and engineering applications.
3. Interpret and analyse data presented in various formats such as tables, graphs, and charts to support statistical analysis and informed decision-making.
4. Apply geometric principles and trigonometric relationships to analyze shapes, angles, and measurements; solve problems involving distances, heights, areas, and volumes; interpret diagrams and spatial data; and use these concepts effectively in mathematical, scientific, and engineering applications.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 1 | | 1 | | 1 | | | | | | 2 |
| CO2 | 3 | 1 | | 1 | | 1 | | | | | | 2 |
| CO3 | 3 | 1 | | 1 | | 1 | | | | | | 2 |
| CO4 | 3 | 1 | | 1 | | 1 | | | | | | 2 |

| Unit1 | Number Systems-II and Algebra | 3 hours |
|---|---|---|
| Types of numbers: (Surds indices, Logarithms, Progressions, Factors, Factorizations, Number of Zeros, Algebraic Identities, Linear and Quadratic Equations. | | |
| **Unit 2** | **Permutation, Combination and Probability** | **3 hours** |
| Set theory, Definition and Introduction of permutation and combination, Fundamental principle of counting by "AND" or "OR" rule, Forming of Numbers, Words, and Team; Problems related to linear and nonlinear arrangement. Application based on Geometry, Match, handshake, Chess board, Whole number, and Natural number solution, etc. Distribution: - Identical & Distinct, Basic concepts, definition and terminology related to probability, Problem based on Dice, Coins, Playing cards, Marbles, etc. Conditional Probability – Bayes theorem. | | |
| **Unit 3** | **Data Interpretation, Data Sufficiency and Statistics** | **3 hours** |
| Data Interpretation: Tables (Simple & Complex), Bar Graphs, Line Graphs, Pie Charts, Histograms, Radar graph, Caselets, Mixed Charts, Missing Data Problems, Comparison & Ratio Analysis, Percentage Change Analysis, Time-Based Data Sets, Data Sufficiency.<br>Statistics: Measures of Central Tendency (Mean, Median, Mode), Measures of Dispersion (Range, Mean Deviation, Variance, Standard Deviation), Averages & Weighted Mean, Quartiles, Deciles & Percentiles. | | |
| **Unit 4** | **Geometry, Mensuration and Trigonometry** | **3 hours** |
| **Geometry**: Basics of Geometry: Lines and Angles, Triangles (Types, Properties, Theorems), Congruence of Triangles, Similarity of Triangles, Quadrilaterals (Parallelogram, Rectangle, Square, Rhombus, Trapezium), Polygons, Circles (Chord, Tangent, Secant, Angles in a Circle). **Mensuration** – 2D Figures (Area & Perimeter), Mensuration – 3D Figures (Surface Area & Volume). **Trigonometry:** Introduction to Trigonometry, Trigonometric Ratios (sin, cos, tan, cosec, sec, cot), Trigonometric Identities, Values of Trigonometric Functions (Standard Angles), Trigonometric Equation, Heights and Distances. | | |
| **Unit 5** | **Flowcharts and Gamified Aptitude** | **3 hours** |
| Number Puzzles, Crypt Arithmetic, Flowcharts, Geo-Sudoku, Digit Challenge, Switch Challenge, Motion Challenge, Grid Challenge. | | |
| | **Total Lecture hours** | **15 hours** |

**Textbook**

1. Booklet provided by the Aptitude Team, Training Division -CRPC, KIET

**Reference Books**

1. "Quantitative Aptitude for Competitive Examinations" by R.S. Aggarwal, S. Chand Publication.
2. "How to Prepare for Quantitative Aptitude for the CAT" by Arun Sharma, TMH Publication.
3. "Magical Book on Quicker Maths" by M. Tyra, Published by BSC Publishing Co. Pvt. Ltd
4. "Quantitative Aptitude Quantum CAT" by Sarvesh K. Verma, Arihant Publication.

**Reference Websites:**

1. https://www.indianbix.com
2. https://www.prepinsta.com
3. https://www.talentsprint.com
4. https://www.careeride.com
5. https://www.smartkeeda.com

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| MSE | CA | ESE | Total Marks |
| - | 25 | 25 | 50 |
| 25 | | | |

| Course Code: AI202B | Course Name: Fundamentals of Cyber Security | L | T | P | C |
|---|---|---|---|---|---|
| | | 3 | 0 | 2 | 4 |

**Pre-requisite: NIL**

**Course Objective:** To enable students to understand and apply cybersecurity concepts, security technologies, digital forensics, and cyber laws to protect computer networks and information systems.

**Course Outcomes:**

CO1: Explain cyber security fundamentals, types of cyber threats, and analyze basic vulnerabilities using ethical hacking principles.
CO2: Apply network security concepts including firewalls, IDS/IPS, VPNs, and analyze network traffic to identify potential security threats.
CO3: Analyze common web and database vulnerabilities and apply secure authentication, encryption, and access control mechanisms.
CO4: Apply identity and access management principles, including multi-factor authentication and role-based access control, to enhance system security.
CO5: Analyze digital evidence using basic forensic techniques and interpret relevant cyber laws in the context of cyber incidents.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 1 | – | 2 | 1 | 2 | – | – | – | 1 |
| CO2 | 3 | 3 | 2 | 2 | 3 | 1 | 1 | – | – | – | 1 |
| CO3 | 3 | 3 | 3 | 2 | 2 | 1 | 2 | – | – | – | 1 |
| CO4 | 3 | 3 | 3 | – | 2 | 1 | 2 | 1 | – | – | 1 |
| CO5 | 3 | 3 | 1 | 3 | 2 | 2 | 3 | – | 1 | – | 2 |

| Unit1 | Introduction to Cyber Security | 15 hours |
|---|---|---|

**Theory:**

Introduction to Cyber Security and Cybercrime; Types of Hackers (White Hat, Grey Hat, Black Hat); Cyber Attack Lifecycle (Reconnaissance, Exploitation, Maintaining Access); Malware Types – Virus, Worm, Trojan, Ransomware, Spyware; Social Engineering Techniques – Phishing and Spear Phishing; Password Attacks – Brute Force and Dictionary Attacks; Vulnerabilities and Exploits; Secure Coding Principles – Input Validation and Least Privilege; Introduction to Ethical Hacking and VAPT.

**Laboratory (Tool: OWASP Juice Shop):**

Lab 1: Exploring web application vulnerabilities and identifying insecure login mechanisms.
Lab 2: Demonstration of basic injection and authentication bypass techniques in a controlled environment.

| Unit 2 | Network Security Fundamentals | 15 hours |
|---|---|---|

**Theory:**

OSI and TCP/IP Models (Security Perspective); Basics of IP Addressing and Port Numbers; Common Network Attacks – DoS, DDoS, Man-in-the-Middle, ARP Spoofing; Firewalls – Packet Filtering and Stateful Inspection; Intrusion Detection and Prevention Systems (IDS/IPS); Virtual Private Networks (VPN) – Basic Concepts; HTTPS and SSL/TLS; Network Hardening Techniques.

**Laboratory (Tool: Wireshark):**

Lab 3: Packet capture and analysis of network protocols (HTTP, DNS, TCP).
Lab 4: Identification of suspicious network activity and password cracking.

| Unit 3 | Web and Database Security | 15 hours |
|---|---|---|

**Theory:**

Web Application Architecture and Client-Server Model; HTTP vs HTTPS; Common Web Vulnerabilities – SQL Injection, Cross-Site Scripting (XSS), Broken Authentication; Secure Authentication Mechanisms; Role-Based Access Control (RBAC); Encryption Fundamentals – Symmetric and Asymmetric Encryption; Hashing Concepts; Secure Password Storage – Salting and Hashing; Backup and Basic Disaster Recovery Concepts.

**Laboratory (Tool: Nmap):**

Lab 5: Network and service scanning to identify open ports and running services.
Lab 6: Basic vulnerability detection using Nmap scripts and interpretation of scan results.

| Unit 4 | Identity, Access Management and Mobile Security | 15 hours |
|---|---|---|

**Theory:**
Identity and Access Management (IAM) Overview; Authentication vs Authorization; Factors of Authentication; Multi-Factor Authentication (MFA); Password Policies and Account Lockout Policies; Privileged Access Management; Introduction to Active Directory; Mobile Security Risks – Malicious Applications, Permission Abuse, Public Wi-Fi Threats; Organizational Security Policies and Best Practices.

**Laboratory (Tool: Google Authenticator / Authy):**
Lab 7: Implementation and testing of Multi-Factor Authentication (MFA) on a demo platform.
Lab 8: Configuration of role-based access control and permission management.

| Unit 5 | Digital Forensics and Cyber Law | 15 hours |
|---|---|---|

**Theory:**
Introduction to Digital Forensics; Types of Digital Evidence; Chain of Custody; Digital Forensics Investigation Process; Log Analysis Basics; Email Header Analysis; Basics of Incident Response; Overview of IT Act 2000 (Key Provisions); Overview of Digital Personal Data Protection Act 2023; Introduction to Intellectual Property – Copyright and Patent Basics.

**Laboratory (Tool: Autopsy):**
Lab 9: Disk image examination and identification of suspicious files.
Lab 10: Recovery of deleted files and preparation of a basic incident investigation report.

| Total Lecture hours | 75 hours |
|---|---|

**Textbooks:**

1. Brooks, C. J., & Grow, C. (2024). Cybersecurity Essentials (2nd ed.). Wiley.
2. Sammons, J. (2024). The Basics of Digital Forensics: The Primer for Getting Started in Digital Forensics (3rd ed.). Syngress/Elsevier.

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| MSE | CA | ESE | Total Marks |
| 80 | 20 | 100 | 200 |
| 100 | | | |

| Course Code: AI403B | Course Name: System Design | L | T | P | C |
|---|---|---|---|---|---|
| | | 3 | 0 | 2 | 4 |

**Pre-requisite:** C++ Programming, OOP Concepts, and Data Structures

**Course objective:** To enable students to analyze, design, and implement scalable software systems using object-oriented design principles, UML, design patterns, and advanced low-level design practices.

**Course Outcomes:**
CO1: Apply object-oriented design principles, SOLID principles, and UML modeling techniques in software design.
CO2: Implement creational and structural design patterns in C++ to solve structured software design problems.
CO3: Design and implement behavioral patterns while evaluating appropriate pattern usage and avoiding anti-patterns.
CO4: Design complete low-level architectures for real-world systems and translate UML diagrams into modular C++ code.
CO5: Apply advanced LLD concepts including concurrency control, memory management, clean architecture, dependency injection, and testable design.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | 3 | 1 | 2 | 1 | 1 | 1 | 1 | 1 | 1 |
| CO2 | 2 | 3 | 3 | 2 | 3 | 1 | 1 | 1 | 1 | 1 | 1 |
| CO3 | 2 | 3 | 3 | 2 | 3 | 1 | 1 | 1 | 1 | 1 | 1 |
| CO4 | 3 | 3 | 3 | 3 | 3 | 1 | 1 | 1 | 2 | 2 | 2 |
| CO5 | 3 | 3 | 3 | 3 | 3 | 1 | 1 | 2 | 2 | 2 | 2 |

| Unit1 | Object-Oriented Design Principles | 15 hours |
|---|---|---|
| Review of OOP concepts in C++, Classes, Objects, Inheritance, Polymorphism, Encapsulation, Abstraction, SOLID Principles with C++ examples (Single Responsibility Principle, Open/Closed Principle, Liskov Substitution Principle, Interface Segregation Principle, Dependency Inversion Principle), DRY, KISS, YAGNI principles, Composition vs Inheritance, Abstract classes and Pure virtual functions in C++, Introduction to UML, Class Diagrams, Object Diagrams, Sequence Diagrams | | |
| **Unit 2** | **Design Patterns Creational & Structural** | **15 hours** |
| Introduction to Design Patterns (GoF patterns overview), Creational Patterns (Singleton Pattern, Factory Method Pattern, Abstract Factory Pattern, Builder Pattern, Prototype Pattern), Structural Patterns (Adapter Pattern, Decorator Pattern, Facade Pattern, Proxy Pattern, Composite Pattern, Bridge Pattern), Implementing each pattern in C++ with real-world analogies | | |
| **Unit 3** | **Design Patterns — Behavioral** | **15 hours** |
| Behavioral Patterns (Observer Pattern, Strategy Pattern, Command Pattern, Iterator Pattern, State Pattern, Template Method Pattern, Chain of Responsibility Pattern, Mediator Pattern, Memento Pattern), Anti-patterns and when NOT to use design patterns, Pattern combinations in real-world systems | | |
| **Unit 4** | **LLD of Real-World Systems** | **15 hours** |
| Approach to solving LLD problems (requirement gathering, identify entities, define relationships, apply patterns), LLD of (Parking Lot System, Library Management System, Elevator/Lift System, Online Bookstore / E-commerce Cart, ATM Machine, Ride Sharing App (Uber/Ola simplified), Class diagram design for each system, Writing clean, modular C++ code for the above, Handling edge cases and extensibility | | |
| **Unit 5** | **Advanced LLD Concepts & Best Practices** | **15 hours** |
| Introduction to Concurrency in LLD — Mutex, Locks, Thread-safe Singleton in C++, Designing with Smart Pointers (unique_ptr, shared_ptr, weak_ptr) for memory-safe LLD, Exception handling in design, Dependency Injection in C++, Introduction to Clean Architecture and Layered Design, Code refactoring techniques, Writing testable code introduction to unit testing in C++ (Google Test basics), LLD Interview approach and whiteboard design techniques | | |
| | **Total Lecture hours** | **75 hours** |

**Textbook:**

1. Dennis, A., Wixom, B. H., & Roth, R. M. Systems Analysis and Design. 8th Edition, Wiley, 2020.
2. Winters, T., Manshreck, T., & Wright, H. (Eds.). Software Engineering at Google: Lessons Learned from Programming Over Time. O'Reilly Media, 2020.

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| **MSE** | **CA** | **ESE** | **Total Marks** |
| **80** | **20** | **100** | **200** |
| **100** | | | |

## Open Elective-I Courses Detail Syllabus – VI Sem

| Course Code: MA202O | Course Name: Reliability Theory and Maintenance | L | T | P | C |
|---|---|---|---|---|---|
| | | 2 | 0 | 0 | 2 |

**Pre-requisite: 10+2**

**Course Objectives:**

1. The objective of this course is to familiarize the graduate engineers with the concept of Reliability.
2. To equip students with techniques for reliability modeling, maintenance planning, and optimization of system availability and maintainability.

Course Outcome:

1. Understand the concept of Reliability of system with failure and distribution function.
2. Apply the concept of reliability on redundant system.
3. Employ the concept of reliability on complex system.
4. Understand the knowledge of Modeling, Maintainability and Availability of the system

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO-PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 2 | 2 | | | | 1 | | | | 2 |
| CO2 | 2 | 2 | 2 | | | | 1 | | | | 2 |
| CO3 | 2 | 2 | 2 | | | | 1 | | | | 2 |
| CO4 | 2 | 2 | 2 | | | | 1 | | | | 2 |

| Unit 1 | Introduction to Reliability | 08 hours |
|---|---|---|

Definition of Reliability, Concept of Failure (Failure, failure rate, hazard rate), Types of failures: Early failure (infant mortality), Random failure, Wear-out failure; Failure density function, Bathtub Curve, Mean time to failure, Mean Time To Repair (MTTR), Mean Time Between Failures (MTBF), Estimation of Reliability function from failure data. Probability Distributions in Reliability: Binomial distribution, Poisson distribution, Exponential distribution, Weibull distribution

| Unit 2 | System Reliability | 08 hours |
|---|---|---|

Configuration of System components: Series Configuration, Parallel Configuration, Mixed configurations, k-out-of-n:G/F Configuration, Definition of Redundancy, Types of Redundancy: Active redundancy and Standby (passive) redundancy; Reliability evaluation of the system (Series systems, Parallel systems, Mixed system); Reliability of k-out-of-n:G/F system, reliability of Standby system (with perfect and imperfect switching).

| Unit 3 | Reliability of Complex Systems | 07 hours |
|---|---|---|

Definition of Complex system, Decomposition method, Conditional Probability approach, Minimal Path set method, Cut set method, Tie set method, Reliability Based Optimization, Introduction to Stochastic FEM

| Unit 4 | Modeling, Maintainability and Availability | 07 hours |
|---|---|---|

Reliability Block Diagrams (RBD), Fault tree analysis (FTA), Life Testing and Estimation: Parameter estimation (MLE method) and Goodness-of-fit tests, Markov models, Maintainability function, Availability function

| | Total hours | 30 hours |
|---|---|---|

**Textbook:**

1: E. Balaguruswamy, “Reliability Engineering”, Tata McGraw Hill.
2: D J Smith, Reliability, Maintainability & Risk, Butterworth-Heinemann

**Reference Books:**

1: Charles E Ebeling, “Reliability and Maintainability Engineering, Tata McGraw Hill
2: R.Billintan and R.N. Allan, "Reliability Evaluation of Engineering and Systems", Plenum Press.
3: K.C. Kapoor& L.R. Lamberson, "Reliability in Engineering and Design", John Wiely and Sons.
4: G.H.Sandler, "System Reliability Engineering", Prentice Hall.

**Mode of Evaluation:**

| MSE | CA | ESE | Total |
|---|---|---|---|
| 40 | 10 | 50 | 100 |

**Continuous Assessment (CA):** Video Assignments/ Assignment/ Quiz/ Project/ Seminar/ Group Discussion/ MOOC Course etc.

| Course Code: EC237O | Course Name: Introduction to Robotic Systems | L | T | P | C |
|---|---|---|---|---|---|
| | | 2 | 0 | 0 | 2 |

**Course Objectives:**

1. To understand fundamentals of robotic systems, components and configurations.
2. To analyze robot kinematics, including forward/inverse kinematics and coordinate transformations.
3. To develop mathematical models of robot manipulators using DH parameters, Jacobian, and dynamic approaches.
4. To apply AI techniques (ANN, RL, Computer Vision) in robotic systems for perception, control, and decision-making.

**Course Outcome:** After completion of the course, the student will be able to

1. Explain and classify different types of robots, components, DOF and configurations.
2. Apply forward and inverse kinematics and perform coordinate transformations using homogeneous matrices.
3. Develop kinematic and dynamic models of robotic manipulators using DH parameters, Jacobian, and dynamic equations, and analyze system behavior.
4. Implement AI-based solutions such as: Object detection using OpenCV & YOLO and ANN-based control and trajectory generation.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 2 | 2 | - | - | 2 | - | - | 2 | 2 | - | 2 |
| CO2 | 2 | 2 | 2 | - | - | 2 | - | - | 2 | 2 | - | 2 |
| CO3 | 3 | 3 | 3 | - | - | 3 | - | - | 3 | 3 | - | 2 |
| CO4 | 3 | 2 | 2 | - | 3 | 3 | - | 3 | 2 | 2 | 3 | 3 |

| Unit 1 | Introduction to Robotic Systems | 7 hours |
|---|---|---|

Fundamentals of Robotics: Types of Robots, Robot Components, Manipulator, Reference Frames, Degrees of Freedom, Joints of a Robot, Robot Configurations, End Effector.

| Unit 2 | Kinematic Transformations | 7 hours |
|---|---|---|

Kinematics Equations: Forward Kinematics & Inverse Kinematics, Workspace

Coordinate Frames and Homogeneous Transformation: Coordinate Transformations, Rotation, Translation, Composition of homogeneous transformations

| Unit 3 | Kinematic Model for Robot Manipulator | 8 hours |
|---|---|---|

Denavit-Hartenberg Procedure & Coordinates Frames, 3-DOF & 6-DOF Manipulator, 4-Axis SCARA Robot

Manipulator Jacobian

Dynamics of Manipulator: Lagrange-Euler approach, Newton-Euler approach, Multiple Degree of Freedom

| Unit 4 | AI-Integrated Robotic Systems | 8 hours |
|---|---|---|

Introduction to AI in Robotics: Role of AI in modern robotics, Difference between Traditional vs Intelligent robots

AI for Robot Perception: Sensor data processing, Basics of computer vision using OpenCV, Object detection

| Total Lecture hours | 30 hours |
|---|---|

**Textbook**

1. Subir Kumar Saha, “Introduction to Robotics”, Tata McGraw-Hill Education, 2014.
2. Richard Paul, Robot Manipulators: Mathematics, Programming and Control, MIT Press, 1981
3. Robert Shilling, Fundamentals of Robotics, Prentice-Hall , 2003
4. Laxmidhar Behera and Indrani Kar, “Intelligent Systems and Control”, Oxford University Press, Nov 2009.

**Reference Books**


1. M. Felix Orlando, Ashish Dutta, Anupam Saxena, Laxmidhar Behera, Tomoya Tamei and Tomohiro Shibata, "Manipulability Analysis of Human Thumb, Index and Middle Finger in Cooperative 3D Rotational Movements of a Small Object", Robotica, vol. 31, pp. 797-809, 2013.
2. M. Felix Orlando, Laxmidhar Behera, Tomayo Tamei, Tomohiro Shibata, Ashish Dutta and Anupam Saxena," On Redundancy Resolution of the Human Thumb, Index and Middle Fingers in Cooperative Object Translation," Robotica, vol. 35, pp. 1992-2017, 2016.


**Mode of Evaluation:**

| MSE | CA | ESE | Total |
|---|---|---|---|
| 40 | 10 | 50 | 100 |

**CA1-** Report Submission on Addverb Technologies Visit**, CA2-**Assignment Based**, CA3-**Attendance

| Course Code: EE214O | Course Name: AIoT for Smart Energy | L | T | P | C |
|---|---|---|---|---|---|
| | | 2 | 0 | 0 | 2 |

**Pre-requisite:** Basic knowledge of Electronics, Programming in Python/C, and Mathematics.

**Course Objectives:**

1. To introduce the fundamentals of Artificial Intelligence of Things (AIoT), its architecture in the context of smart energy systems and to design IoT sensor networks for real-time energy monitoring, data acquisition, edge-level processing.
2. To develop the ability to apply machine learning and AI algorithms for energy demand forecasting, anomaly detection, renewable energy prediction and smart grid technologies, cloud platforms, and the future roadmap of AIoT in sustainable energy.

**Course Outcome:** After completion of the course, the student will be able to

1. Understand the fundamentals of AIoT architecture, communication protocols, and their role in the smart energy ecosystem.
2. Design and analyze IoT sensor networks for real-time energy monitoring and data acquisition.
3. Apply machine learning and AI algorithms to predict energy demand and detect anomalies in energy systems.
4. Evaluate smart grid technologies, cloud platforms, renewable energy integration, and AIoT-based optimization strategies.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 2 | 2 | 2 | 2 | - | - | - | - | - | 2 |
| CO2 | 3 | 2 | 2 | 2 | 3 | - | - | - | - | - | 3 |
| CO3 | 3 | 2 | 2 | 2 | 3 | - | - | - | - | - | 3 |
| CO4 | 3 | 2 | 2 | 2 | 3 | - | - | - | - | - | 3 |

**Unit 1 Foundations of AIoT and Smart Energy** — **7 hours**

Overview of AIoT: convergence of Artificial Intelligence and Internet of Things. AIoT system architecture: Perception, Network, Platform and Application layers. Introduction to smart energy systems, smart meters and Advanced Metering Infrastructure (AMI). Key energy metrics: kWh, Power Factor and Load Factor. Communication protocols: MQTT, Zigbee and LoRaWAN. Edge computing versus cloud computing in energy applications.

**Unit 2 IoT Sensors, Devices and Connectivity** — **8 hours**

Energy sensors: current transformers, voltage sensors, irradiance sensors and smart plugs. Microcontroller platforms: Arduino and ESP32 for energy monitoring. Wireless communication: Zigbee and Bluetooth for short range, LoRaWAN for long range, NB-IoT for smart grid applications. Introduction to Edge AI: TinyML and TensorFlow Lite for on-device processing. Basic data pre-processing: noise filtering, handling missing values and time-series data characteristics.

| **Unit 3 Artificial Intelligence for Energy Management** | **8 hours** |
|---|---|

Overview of supervised and unsupervised machine learning with energy examples. Energy demand forecasting using Linear Regression, Random Forest and LSTM networks. Anomaly detection for electricity theft and equipment faults using Isolation Forest and Autoencoders. Solar and wind power prediction using weather data and ML models. Introduction to Reinforcement Learning for smart thermostat and battery scheduling optimization.

| **Unit 4 Smart Grids, Cloud Platforms and Future Applications** | **7 hours** |
|---|---|

Smart grid architecture: evolution from traditional grid to smart grid, demand response and Virtual Power Plants. Cloud platforms: AWS IoT Core, Google Cloud IoT and open-source tools such as Node-RED and ThingSpeak. AIoT for building energy management and EV smart charging. Introduction to Vehicle-to-Grid (V2G) technology. Emerging trends: Federated Learning for energy data privacy, Blockchain for peer-to-peer energy trading and role of AIoT in achieving Net Zero targets.

| **Total Lecture Hours** | **30 hours** |
|---|---|

**Textbook:**


1. A. Bahga and V. Madisetti, *Internet of Things: A Hands-On Approach,* VPT, 2014.
2. F. Chollet, *Deep Learning with Python,* Manning Publications, 2021.
3. J. Ekanayake, K. Liyanage, J. Wu, A. Yokoyama, N. Jenkins, *Smart Grid: Technology and Applications,* Wiley, 2012.
4. A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow,* 3rd ed., O'Reilly Media, 2022.


**Reference Books:**


1.S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach,* 4th ed., Pearson, 2020.
2.R. H. Shumway and D. S. Stoffer, *Time Series Analysis and Its Applications,* Springer, 2017.
3.M. Schwartz, *Internet of Things with ESP8266,* Packt Publishing, 2016.
4.P. J. Sadalage and M. Fowler, *NoSQL Distilled,* Addison-Wesley, 2012.


**Mode of Evaluation:**

| MSE | CA | ESE | Total |
|---|---|---|---|
| 40 | 10 | 50 | 100 |

| **Course Code: ME229O** | **Course Name: Advanced Engineering Materials** | **L** | **T** | **P** | **C** |
|---|---|---|---|---|---|
| | | **2** | **0** | **0** | **2** |

**Pre-requisite:** Basic knowledge of Physics and Chemistry

**Course Objectives:**

1. To introduce students to the classification and properties of advanced engineering materials.
2. To develop understanding of modern materials used in aerospace, biomedical, electronics, and energy applications.
3. To familiarize students with Semiconductors,smart materials, nanomaterials, and composite materials.
4. To enable students to correlate material properties with engineering applications and sustainability requirements.

**Course Outcome:** After completion of the course, the student will be able to

1. Explain the structure, properties, and classification of advanced engineering materials.
2. Analyze the applications of composite, smart, nano, and biomaterials in modern engineering systems.
3. Select suitable advanced materials for specific industrial and technological applications.
4. Understand sustainable and emerging material technologies for future engineering challenges.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 2 | - | - | - | - | - | - | - | - | 2 |
| CO2 | 2 | 3 | 2 | 1 | 1 | - | - | - | - | - | 2 |
| CO3 | 2 | 3 | 3 | 1 | 2 | - | - | - | - | - | 2 |
| CO4 | 2 | 2 | 1 | - | - | 1 | 1 | - | - | - | 3 |

| Unit 1 | Composite materials | 7 hours |
|---|---|---|

Introduction to composite materials and their Classification. Fiber reinforced composites (FRP, CFRP, GFRP) Metal Matrix Composites (MMC), Ceramic Matrix Composites (CMC), Manufacturing methods and applications Properties and applications of composites in automotive and aerospace sectors.

| Unit 2 | Semiconductor and Smart Materials | 8 hours |
|---|---|---|

Introduction to conductors, semiconductors, and insulators, Crystal structure of semiconductors, Energy band theory, Intrinsic and extrinsic semiconductors, Carrier concentration, mobility, conductivity, Fermi level concept Smart materials: shape memory alloys, piezoelectric materials, magnetostrictive materials, and electroactive materials. Applications of smart materials in sensors and actuators.

| Unit 3 | Nanomaterials and Biomaterials | 8 hours |
|---|---|---|

Introduction to nanomaterials and nanotechnology. Synthesis and characterization of nanomaterials. Carbon nanotubes, graphene, nanoparticles, and Nano coatings. Applications of nanomaterials in electronics, energy storage, and environmental engineering. Biomaterials: metallic, ceramic, polymeric, and composite biomaterials. Applications in medical implants and tissue engineering.

| Unit 4 | Advanced Functional Materials and Sustainable Materials | 7 hours |
|---|---|---|

Semiconductors and electronic materials. Superconducting materials and optical materials. Materials for renewable energy systems such as solar cells, batteries, and fuel cells. Eco-friendly and sustainable materials. Recycling and green processing of materials. Recent trends and future scope of advanced engineering materials.

| | Total Lecture Hours | 30 hours |
|---|---|---|

**Textbook:**

1. Callister, W.D. and Rethwisch, D.G., Materials Science and Engineering: An Introduction, Wiley India.
2. Raghavan, V., Materials Science and Engineering, Prentice Hall India.
3. Donald R. Askeland and Wendelin J. Wright, The Science and Engineering of Materials, Cengage Learning.
4. William F. Smith and Javad Hashemi, Foundations of Materials Science and Engineering, McGraw Hill.

**Reference Books:**

1. Charles P. Poole Jr. and Frank J. Owens, Introduction to Nanotechnology, Wiley.
2. M.F. Ashby, H. Shercliff, and D. Cebon, Materials: Engineering, Science, Processing and Design.
3. S.O. Kasap, Principles of Electronic Materials and Devices, McGraw Hill.
4. K.K. Chawla, Composite Materials: Science and Engineering, Springer.

**Mode of Evaluation**

| MSE | CA | ESE | Total |
|---|---|---|---|
| 40 | 10 | 50 | 100 |

| Course Code: MS203O | Theory Course Name: Business Strategy | L | T | P | C |
|---|---|---|---|---|---|
| | | 2 | 0 | 0 | 2 |

**Pre-requisite:** NIL

**Course Objectives**

1. To develop strategic thinking from a business perspective
2. To analyse industry dynamics and strategic positioning.
3. To enable students to design business strategies in technology-driven environments.

**Course Outcome (COs)**

| CO No. | Statement of Course Outcome |
|---|---|
| After the completion of the course, the student will be able to: | |
| CO 1 | **Apply** strategic tools to business scenarios |
| CO 2 | **Analyze** competitive environments and firm positioning |

| | |
|---|---|
| **CO 3** | **Evaluate** strategic choices and business models |
| **CO 4** | **Assess** the impact of innovation and emerging technologies on business strategy. |

**CO-PO Mapping (scale 1: low, 2: Medium, 3: High)**

| Course Outcomes (COs) | Programme Outcomes (POs) | | | | | Additional Programme Outcomes (APOs) | |
|---|---|---|---|---|---|---|---|
| | PO1 | PO2 | PO3 | PO4 | PO5 | APO 1 | APO 2 |
| CO 1 | 1 | 2 | 2 | 2 | 2 | 3 | 3 |
| CO 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| CO 3 | 2 | 3 | 3 | 3 | 2 | 3 | 3 |
| CO 4 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |

**DETAILED SYLLABUS**

| Unit 1 | Foundations of Business Strategy | Hours 7 |
|---|---|---|

Strategy vs Business Model, Levels of strategy, Strategic management process, Concept of Competitive Advantage, Stakeholders in Business Strategy,

| Unit 2 | Industry and Competitive Analysis | Hours 8 |
|---|---|---|

Industry Structure and Dynamics, Porter's Five Forces Model, Competitor Analysis , Market Segmentation and Targeting (basic concepts), Strategic Positioning, Tools: SWOT Analysis, PESTLE

| Unit 3 | Strategic Choices and Business Models | Hours 8 |
|---|---|---|

Generic Strategies: Cost Leadership vs Differentiation, Focus Strategy (Niche markets), Growth Strategies: Expansion, Diversification, Innovation, Role of Technology and AI in shaping business strategy.

| Unit4 | Strategy Implementation & Contemporary Issues | Hours 7 |
|---|---|---|

Strategy Implementation: Organizational Structure and Culture, Leadership and Change Management, Resource Allocation. Contemporary Strategic Issues: Digital Transformation, Artificial Intelligence in Strategy, Sustainability and Corporate Social Responsibility (CSR), Ethics in Strategic Decision-Making.

| | |
|---|---|
| **Total Lecture hours** | **30 hours** |

**Textbook**

1. David & David – *Strategic Management: Concepts and Cases*
2. Michael Porter – *Competitive Strategy*
3. Osterwalder & Pigneur – *Business Model Generation*
4. Selected case studies (Harvard/industry-based simplified versions)

**Reference Books**

1. https://www.ebookbou.edu.bd/Books/Text/SOB/MBA/mba_3311/Unit-07.pdf
2. https://nptel.ac.in/courses/110/108/110108047/
3. https://swayam.gov.in/nd2_imb19_mg08/preview

**Mode of Evaluation**

**Continuous Assessment (CA): Case Presentation/ Activities/ Assignment / Quiz / Project/ MOOC etc.**

| MSE | CA | ESE | Total |
|---|---|---|---|
| 40 | 10 | 50 | 100 |

# Professional Elective-IV Courses Detail Syllabus – VI Sem

| Course Code: AI312E | Course Name: Quantum Computing Essentials | L | T | P | C |
|---|---|---|---|---|---|
| | | 1 | 0 | 2 | 2 |

**Pre-requisite:** Programming Fundamentals , Data Structures and Algorithms , Probability and Statistics ,Basic Linear Algebra , Computer Organization and Digital Logic (basic understanding)

**Course Objective:** To enable students to understand and apply the principles of quantum computing for designing quantum circuits, implementing quantum algorithms, and analyzing quantum hardware.

**Course outcomes:**

CO1: Understand fundamentals of quantum computing and qubit systems.

CO2: Apply quantum gates and design quantum circuits.

CO3: Implement basic quantum algorithms.

CO4: Analyse quantum hardware, noise, and error models.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 1 | 1 | - | 1 | - | 1 | - | 1 | - | 2 |
| CO2 | 2 | 3 | 3 | 2 | 2 | 1 | 1 | 1 | 1 | 1 | 2 |
| CO3 | 2 | 3 | 3 | 3 | 2 | 1 | 1 | 2 | 2 | 1 | 2 |
| CO4 | 2 | 2 | 3 | 3 | 3 | 1 | 1 | 1 | 2 | 2 | 2 |

| Unit 1 | Introduction to Quantum Computing & Qubits | 10 hours |
|---|---|---|

**Topics:**
- Introduction to Quantum Computing
- Classical vs Quantum Computing
- Qubits and Superposition
- Bloch Sphere Representation
- Quantum Measurement

**Hands-on:**
- Install Python & Qiskit
- Statevector simulation
- Bloch sphere visualization

**Case Study:**
Quantum vs Classical computation (speed comparison example)

| Unit 2 | Quantum Gates & Circuits | 11 hours |
|---|---|---|

**Topics:**
- Single-Qubit Gates (X, Y, Z, H)
- Multi-Qubit Gates (CNOT)
- Quantum Entanglement & Bell States
- Quantum Circuit Design
- Measurement in Circuits

**Hands-on:**
- Implement basic gates using Qiskit
- Create Bell states
- Build multi-qubit circuits

| Unit 3 | Quantum Algorithms | 12 hours |
|---|---|---|

**Topics:**
- Quantum Hardware Basics

• Noise and Decoherence
• Quantum Error Analysis
• Quantum Cryptography (BB84 Protocol)
• Applications: Machine Learning & Optimization

**Hands-on:**
• Run circuits on real quantum hardware
• Compare noisy vs ideal simulations
• Simulate BB84 protocol

**Case Study:**
Build a small quantum application: Quantum cryptography OR Optimization / search problem

| **Unit 4** | Quantum Hardware, Noise & Applications | **12 hours** |
|---|---|---|

**Topics:**
• Quantum Hardware Basics
• Noise and Decoherence
• Quantum Error Analysis
• Quantum Cryptography (BB84 Protocol)
• Applications: Machine Learning & Optimization

**Hands-on:**
• Run circuits on real quantum hardware
• Compare noisy vs ideal simulations
• Simulate BB84 protocol

**Case Study:**
Build a small quantum application: Quantum cryptography OR Optimization / search problem

| | **Total Lecture Hours** | **45 hours** |
|---|---|---|

**Textbooks:**

1. **Thomas G. Wong, *Introduction to Classical and Quantum Computing*, Rooted Grove, 2022. Online Available: https://www.thomaswong.net/introduction-to-classical-and-quantum-computing-1e4p.pdf**
2. **Kaiser, S., & Granade, C. (2023). *Learn Quantum Computing with Python and IBM Quantum* (2nd ed.). Packt Publishing.**

**Mode of Evaluation:**

| **Evaluation Scheme** | | | |
|---|---|---|---|
| **MSE** | **CA** | **ESE** | **Total Marks** |
| **40** | **10** | **50** | **100** |
| **50** | | | |

| **Course Code: AI313E** | **Course Name: Ethical AI** | **L** | **T** | **P** | **C** |
|---|---|---|---|---|---|
| | | **1** | **0** | **2** | **2** |

**Pre-requisite: Basic understanding of AI/ML concepts**

**Course Objectives:** To enable students to understand and apply Responsible AI principles for developing fair, transparent, accountable, and inclusive AI systems in compliance with Indian governance and regulatory frameworks.

**Course outcomes:**

CO1: Explain the foundations of AI ethics grounded in Indian constitutional values, NITI Aayog responsible AI principles, and landmark Indian case studies.

CO2: Identify, measure, and mitigate bias and fairness issues in AI systems with focus on Indian socio-demographic contexts and DPDP Act 2023 compliance.

CO3: Evaluate accountability, explainability, and governance of AI using MeitY India AI Governance Guidelines, NITI Aayog frameworks, and XAI tools.

CO4: Design responsible AI policies addressing generative AI risks, deepfake threats, IndiaAI Mission pillars, and inclusion of marginalised communities.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 2 | 1 | 1 | - | 1 | - | 1 | - | 1 | - | 2 |
| CO2 | 2 | 3 | 3 | 2 | 2 | 1 | 1 | 2 | 1 | 1 | 1 |
| CO3 | 2 | 3 | 2 | 3 | 2 | 2 | 2 | 1 | 2 | 1 | 1 |
| CO4 | 2 | 3 | 3 | 3 | 3 | 1 | 1 | 1 | 2 | 2 | 1 |

| Unit 1 | Foundations of AI Ethics & Bias in the Indian Context | 12 hours |
|---|---|---|

**Topics:**

History and context of AI ethics in India: Aadhaar biometrics controversy, predictive policing tools, CCTV surveillance in public spaces
Core ethical principles in Indian context: fairness, transparency, accountability grounded in Indian Constitutional values
NITI Aayog Principles for Responsible AI (2021) — seven pillars: safety, inclusivity, equality, privacy, transparency, accountability, human values
Stakeholder mapping: government, private sector, civil society, marginalised communities in Indian AI deployments
Landmark case studies: Aadhaar-linked exclusions, social credit-like welfare algorithms, automated hiring in Indian IT sector
Types of algorithmic bias in Indian context: caste-based, religion-based, gender-based, linguistic bias in NLP models for Indian languages
Fairness metrics: demographic parity, equalised odds, calibration — applied to Indian judicial, recruitment, and credit-scoring datasets
Underrepresentation of Indian languages, dialects, and demographic groups in global ML training datasets
Bias detection tools: IBM AI Fairness 360, Microsoft Fairlearn — applied to India-specific datasets (loan approvals, job screening)

**Hands-on:**

Case Study Activity: Analyse the Aadhaar biometric authentication system — map all stakeholders, identify harms for excluded groups, and propose one ethical safeguard per harm. Compare with COMPAS (USA) to draw global parallels.
Bias Audit Lab: Using AI Fairness 360, audit a publicly available Indian credit scoring or recruitment dataset. Compare error rates across gender, caste-proxy, and regional attributes. Write a bias report with two mitigation recommendations.
Dataset Analysis: Examine a government dataset (PM-KISAN beneficiary data or NREGA payment records) for demographic representation gaps. Document underrepresented groups and propose corrective data collection strategies.

| Unit 2 | Privacy, Consent & Data Rights — DPDP Act 2023 & Indian Framework | 10 hours |
|---|---|---|

**Topics:**

India's Digital Personal Data Protection (DPDP) Act 2023: data fiduciaries, consent architecture, significant data fiduciaries, Data Protection Board
Aadhaar Act and data minimisation: purpose limitation, storage restriction, and consent mechanisms in India's national identity infrastructure
Surveillance and privacy: UPI transaction monitoring, telecom surveillance, facial recognition in Indian public infrastructure
Privacy-by-design in Indian AI systems: k-anonymity, differential privacy applied to Indian census and health data
Fairness, accountability, transparency and ethics in Indian algorithmic systems (Dr. Abhijnan Chakraborty, Dr. Saptarshi Ghosh)

**Hands-on:**

DPDP Compliance Workshop: Select an Indian fintech or edtech application. Map its data flow — collection, processing, storage, sharing. Identify at least three DPDP Act 2023 compliance gaps. Redesign the consent flow and data handling to meet DPDP standards. Present a before/after comparison.
Technical Lab: Apply differential privacy on an anonymised NREGA or health survey dataset using Google's DP Library. Measure utility-privacy trade-offs at different epsilon values. Reflect on acceptable privacy budgets for Indian government data systems.

| Unit 3 | Accountability, Transparency & Governance — MeitY & NITI Aayog | 12 hours |
|---|---|---|

**Topics:**

Responsible and Safe AI Systems — metrics and tools for responsible AI, LIME/SHAP/GradCAM, adversarial testing, audit mechanisms

India AI Governance Guidelines: risk-tiered governance, whole-of-government approach, AI Safety Institute (AISI), Operationalising Responsible AI: regulatory intervention, capacity building, ethics-by-design in Indian public-sector AI
Explainability in Indian judicial and administrative AI: transparency requirements for automated decision-making in courts, welfare schemes, and policing
Red-teaming Indian AI deployments: adversarial testing of Hindi/regional-language NLP models, facial recognition bias in diverse Indian demographics
NASSCOM AI Ethics & Responsible AI Framework for Indian Industry (2023): governance checklists, audit frameworks for Indian enterprises

**Hands-on:**

Model Card Assignment: Write a model card for a pre-trained Indian language NLP model (e.g., AI4Bharat IndicBERT). Include intended use, out-of-scope uses, multilingual performance across Indian languages, demographic fairness evaluation, and ethical considerations.
SHAP Analysis Lab & Red-Team Report: Run SHAP explainability analysis on a credit or loan approval ML model trained on Indian data. Draft a red-team report identifying top three failure modes specific to Indian socio-economic contexts and recommend both technical and governance mitigations.

| **Unit 4** | **Future Risks, IndiaAI Mission & Responsible Innovation** | **11 hours** |
|---|---|---|

**Topics:**

Generative AI risks in Indian context: deepfakes in Indian elections, synthetic media in regional languages, LLM hallucinations in Hindi/Tamil/Telugu content
Autonomous systems in India: AI in agriculture (crop advisory drones), judicial AI (SUPACE — Supreme Court AI Portal), and predictive policing
IndiaAI Mission: seven pillars — compute, datasets, innovation, startups, skills, safety, sustainability
Value-sensitive design for India: inclusion of marginalised communities, linguistic diversity, rural accessibility, and low-digital-literacy populations in AI design
Responsible AI policy design: DPDP Act compliance, bias mitigation, NITI Aayog principles alignment, transparency and explainability, MeitY governance adherence

**Hands-on:**

Capstone Policy Brief: Design a comprehensive Responsible AI policy for an Indian government use case (Aadhaar-linked welfare delivery, SUPACE judicial AI, digital agriculture advisory, or healthcare diagnostics in rural India). Address DPDP Act compliance, bias mitigation, NITI Aayog principles alignment, transparency, MeitY governance, and inclusion of marginalised communities. Present to a mock AI Ethics Board.
Deepfake & Misinformation Lab: Analyse synthetic media samples targeting Indian political or regional content. Use an open-source deepfake detection tool. Propose platform-level governance interventions aligned with India's IT (Intermediary Guidelines) Rules 2021.

| | **Total Lecture Hours** | **45 hours** |
|---|---|---|

**Textbooks & Indian Reference Materials:**

1. NITI Aayog — Responsible AI for All: Principles for Responsible AI (Part I & II, 2021)
2. NITI Aayog — National Strategy for Artificial Intelligence (#AIforAll, 2018)
3. MeitY — India AI Governance Guidelines (2025)
4. NPTEL — Responsible and Safe AI Systems (IIT Madras / IIIT Hyderabad, 2025) — Prof. PK Kumaraguru, Prof. B. Ravindran, Prof. Arun Rajkumar
5. Shoshana Zuboff — The Age of Surveillance Capitalism (PublicAffairs, 2019)
6. Usha Ramanathan — Privacy, Surveillance, and the Aadhaar Act: Articles and Legal Commentary (IJLS, 2018)
7. NASSCOM — Responsible AI: Operationalising the Framework for Indian Industry (2023)

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| **MSE** | **CA** | **ESE** | **Total Marks** |
| **40** | **10** | **50** | **100** |
| **50** | | | |

| Course Code: AI314E | Course Name: Mobile Application & Development | L | T | P | C |
|---|---|---|---|---|---|
| | | 1 | 0 | 2 | 2 |

**Pre-requisite: JAVA Programming**

**Course Objective:** To enable students to understand and develop Android mobile applications using core Android components, user interface design, and application deployment techniques.

**Course outcomes:**
CO1: Understand the fundamentals of mobile applications and the Android development environment.
CO2: Develop simple Android applications using core components.
CO3: Design user interfaces and implement basic data storage and multimedia features in Android applications.
CO4: Build, test, and deploy basic Android applications.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 1 | - | 1 | - | - | - | - | - | - | 1 |
| CO2 | 2 | 2 | 3 | 2 | 3 | - | - | - | 1 | 1 | 2 |
| CO3 | 2 | 2 | 3 | 2 | 2 | - | - | - | 1 | 1 | 1 |
| CO4 | 2 | 1 | 2 | 2 | 3 | - | - | - | 2 | 1 | 2 |

| Unit 1 | Introduction to Mobile Applications and Android Platform | 10 hours |
|---|---|---|

**Theory**
Introduction to mobile applications: definition, features, benefits, and application areas. Types of mobile apps: Native, Hybrid, and Web apps. Introduction to Android and its features. Basic Android architecture (simple overview). Android application structure. Mobile application development life cycle (basic understanding)

**Lab / Hands-on**

- Installation of Android Studio
- Creating first Android project
- Running the app on Emulator
- Understand project files (MainActivity, layout file, manifest)

| Unit 2 | Android Application Components and User Interface Design | 12 hours |
|---|---|---|

**Theory**
Introduction to Android application components. Activities and Activity lifecycle (basic idea). Intents: explicit and implicit intents. Android Manifest file (purpose and structure). Introduction to Android UI. Layouts: Linear Layout and Constraint Layout. UI Components: TextView, EditText, Button, ImageView. Event handling (Button click, user interaction)

**Lab / Hands-on**

- Create an app with multiple screens
- Navigate between activities using Intent
- Pass data between activities
- Design login and registration screens
- Handle button click events and display user input

| Unit 3 | Data Storage, Multimedia and Resources in Android | 11 hours |
|---|---|---|

**Theory**
Data storage in Android (Internal storage & Shared Preferences). Introduction to SQLite (concept only). Managing application resources. Playing multimedia in Android (audio, images, video)

**Lab / Hands-on**

- Save user data using Shared Preferences
- Create a simple note-saving app
- Display images from resources
- Play an audio file inside an Android app

| Unit 4 | Basic Advanced Features, Testing and Deployment | 12 hours |
|---|---|---|

**Theory**

Introduction to location services (basic idea). Introduction to networking in Android (simple concept).Testing Android applications (overview). Introduction to app deployment and APK generation. Overview of publishing apps on Google Play Store

**Lab / Hands-on**

- Create an app that opens Google Maps
- Fetch simple data from the internet (using URL/API)
- Test the app on Emulator
- Generate APK file

| | |
|---|---|
| **Total Lecture Hours** | **45 hours** |

**Textbooks:**


1. Griffiths, D., & Griffiths, D. Head First Android Development: A Learner's Guide to Building Android Apps. 3rd Edition, O'Reilly Media, 2024.
2. Lee, N. Android Studio Hedgehog Essentials – Kotlin Edition. Payload Media, 2024.
3. Jemerov, D., & Isakova, S. Kotlin in Action. 2nd Edition, Manning Publications, 2024.


**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| **MSE** | **CA** | **ESE** | **Total Marks** |
| **40** | **10** | **50** | **100** |
| **50** | | | |

| **Course Code: AI315E** | **Course Name: System Testing** | **L** | **T** | **P** | **C** |
|---|---|---|---|---|---|
| | | **1** | **0** | **2** | **2** |

**Pre-requisite: Software Engineering Concepts**

**Course Objective:** To enable students to apply software testing techniques and automation tools to ensure software quality and reliability.

**Course Outcomes:**

**CO1:** Apply software testing knowledge and engineering methods.
**CO2:** Design and conduct a software test process for a software testing project
**CO3:** Identify the needs of software test automation, and define and develop a test tool to support test automation.
**CO4:** Design test cases and Understand Test Automation Tools.

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **CO1** | 2 | 2 | 1 | - | 1 | - | 1 | - | 1 | - | 1 |
| **CO2** | 2 | 3 | 3 | 2 | 2 | - | 1 | 2 | 2 | 2 | 1 |
| **CO3** | 2 | 3 | 3 | 2 | 2 | - | 2 | 2 | 2 | 2 | 1 |
| **CO4** | 2 | 3 | 3 | 2 | 3 | - | 2 | 3 | 3 | 2 | 1 |

| **Unit 1** | **Introduction** | **11 hours** |
|---|---|---|

Definition and importance of software testing, Objectives and principles of testing, Verification vs. Validation, Software development life cycle (SDLC) and testing, Testing lifecycle (STLC), Errors, Faults, Failures — terminology, costs of defects, Types of testing: Static vs. Dynamic, Black-box, White-box, and Grey-box testing, Manual vs. Automated testing, Levels of testing: Unit → Integration → System → Acceptance

**Hands on:**

Perform Black box Testing on Sample Application
Perform White box Testing on Sample Application

| Unit 2 | System Testing | 12 hours |
|---|---|---|

Definition and scope of system testing, Objectives and entry/exit criteria, System testing vs. other testing levels, System testing environment setup, Roles and responsibilities in system testing, Types of System Testing- Functional Testing, Regression Testing, Smoke Testing, Sanity Testing, Integration Testing, End-to-End Testing, User Interface (UI) Testing, Database Testing, API Testing. Non Functional Testing- Performance Testing, Load Testing, Stress Testing, Volume Testing, Security Testing, Usability Testing, Compatibility Testing.

**Hands on:**

Install JMeter and set up
Create test plan
Perform performance testing and load testing on sample http requests
Analyze performance report

| Unit 3 | Test Design Techniques | 11 hours |
|---|---|---|

Black-box techniques -Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, State Transition Testing, Use Case Testing, White-box techniques - Statement Coverage, Branch/Decision Coverage, Path Coverage, Condition Coverage, Experience-based techniques - Exploratory Testing, Error Guessing, Test planning process, Test estimation techniques, Risk-based testing, Test metrics and measurement

**Hands on:**

Design test cases using Equivalence Partitioning
Design test cases using Boundary Value Analysis
Apply Decision Table Testing

| Unit 4 | Test Case Design and Execution | 11 hours |
|---|---|---|

Writing effective test cases, Test case components (ID, description, steps, expected result), Test suite, Test plan, Test script, Test data preparation, Requirement Traceability matrix (RTM), Test execution process, Defect reporting and tracking, Defect management lifecycle, Bug lifecycle (New → Open → Fixed → Closed), Need for automation, Automation vs. manual testing, Automation frameworks (Data-driven, Keyword-driven, Hybrid, BDD), Popular tools - Selenium, JUnit, Postman, JMeter, LoadRunner, Test management tools - JIRA, TestRail, Version control-Git, Environment setup

**Hands on:**

Create Test suite for sample application
Create Test script for sample application
Install and setup Selenium
Perform Functional and UI testing on Selenium
Set up Git and Run basic command for version control (e.g.- pull,push,commit)

**Total Hours: 45 hours**

**Textbooks:**

1. Yogesh Singh, "Software Testing", Cambridge University Press, New York, 2012
2.Marc Roper, "Software Testing", McGraw-Hill Book Co., London, 1994.
3.Glenford J. Myers, Corey Sandler, Tom Badgett, "The Art of Software Testing", 3rd Edition, Wiley, 2011

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| MSE | CA | ESE | Total Marks |
| 40 | 10 | 50 | 100 |
| 50 | | | |

# Practical Courses Detail Syllabus – VI SEM

| Course Code: HS206P | Course Name: Soft Skills for Employability - 2 | L | T | P | C |
|---|---|---|---|---|---|
| | | 0 | 0 | 2 | 1 |

**Pre-requisite:** Successful completion of Soft Skills Essentials I & II and Employability Skills – I (Verbal Ability).

**Course Objectives:**
To strengthen students' placement preparedness and develop employability skills by enhancing creativity, critical thinking, resume development, resume preparation, group discussion skills, and interview techniques.

**Course Outcome:** After completion of the lab, the student will be able to
CO1: Demonstrate interpersonal effectiveness through handling group discussions and professional presence through visumes.
CO2: Justify their resumes through effective professional presentation skills.
CO3: Apply interview techniques and feedback to successfully handle employment interviews.
CO4: Develop a professional digital identity by utilizing AI-enabled career tools, networking platforms, psychometric insights, and employability strategies to enhance career readiness.

**CO-PO Mapping (scale 1: low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | - | - | - | - | - | 1 | - | 2 | 3 | - | 2 |
| CO2 | - | - | - | - | - | - | - | - | 3 | - | 2 |
| CO3 | - | - | - | - | - | - | - | 1 | 3 | - | 2 |
| CO4 | - | - | - | - | - | 1 | - | - | 2 | - | 2 |

**List of Activities:**

1. LLM Enabled Professional Self-Introductions using Mind-Mapping
2. Employability & Placement Process Overview
3. Resume Review Lab – Peer Review & Faculty Feedback
4. Group Discussion Practice Sessions
5. Personal Branding for Career Development – LinkedIn, AlmaConnect, E-Portfolio
6. Psychometric Test (MBTI / Locus of Control)
7. Preparation for Job Interviews – 4 Ts (Types, Tips, Tricks & Techniques)
8. Pair & Peer Interview Practice
9. Practising HR, Behavioural and Situational Questions
10. Handling HR Questions (Extempore)
11. Company Specific & Profile Specific Interviews (AI Tools)
12. Online Spoken English Test

**Total Laboratory Hours: 30 hours**

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| MSE | CA | ESE | Total Marks |
| - | 50 | - | 50 |
| 50 | | | |

| Course Code: CS317P | Course Name: Problem Solving Lab-2 | L | T | P | C |
|---|---|---|---|---|---|
| | | 0 | 0 | 2 | 1 |

**Pre-requisite: Knowledge of Data Structure**

**Course Objectives:**

1. To familiarize students with the basic and advance of Problem Solving.
2. To learn modern techniques to solve DSA Problems.

**Course Outcomes:** After completion of the lab, the student will be able to

1.Apply data-driven and combinatorial problem-solving techniques
2.Design solutions using Dynamic Programming
3.Develop solutions using trees and graph algorithms
4. Integrate advanced techniques to solve complex problems

**CO-PO Mapping (Scale 1: Low, 2: Medium, 3: High)**

| CO -PO Mapping | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| CO1 | 3 | 3 | 2 | 1 | 2 | - | - | - | - | - | 1 |
| CO2 | 3 | 3 | 3 | 1 | 2 | - | - | - | - | - | 1 |
| CO3 | 3 | 3 | 3 | 2 | 2 | - | - | - | - | - | 1 |
| CO4 | 2 | 3 | 3 | 2 | 3 | - | - | 2 | 1 | 1 | 2 |

**List of Experiments:**

1. Hashing Basics- 1. Frequency counting 2. First non-repeating character 3. Two sum using hashing
2. Hash Map + Sets Advanced -1. Group anagrams 2. Longest consecutive sequence 3. Subarray sum = k
3. Prefix Sum / Difference Array - 1. Range sum queries 2. Rainwater trapping 3.Kadane's algorithm
4. Recursion +Backtracking Advanced- 1. Sudoku solver 2. Word search in grid 3. Combination sum
5. Dynamic Programming– I (Introduction) - 1 . Fibonacci DP 2. Climbing stairs 3. Min cost climbing stairs
6. DP – II (1D DP) 1. House robber 2. Decode ways 3. Partition equal subset sum
7. DP – III (2D DP) 1. Longest common subsequence 2. Edit distance 3. 0/1 knapsack
8. DP – IV (Advanced) 1. Matrix chain multiplication 2. Palindromic subsequence 3. LIS
9. Trees Basics(Conceptual Only)- 1. Preorder traversal 2. Height of tree 3. BFS on tree
10. Binary Search Tree Techniques - 1. Validate BST 2. LCA 3. Kth smallest element
11. Graphs – BFS/DFS 1. Number of islands 2. Bipartite graph check 3. Connected components
12. Graphs – Shortest Path 1. Dijkstra 2. Bellman–Ford 3. Shortest path in grid
13. Graphs – Advanced 1 . Topological sort 2. Course schedule 3. Detect cycle in directed graph
14. Mixed Techniques / DSA Integration - 1. Sliding window + hashing problem 2. Graph + BFS+ DP hybrid 3. String + DP complex case
15. Final Mini Project 1. 20-question problem sheet 2. Contest style evaluation 3. Full pattern-based lab exam

**Total Laboratory Hours: 30 hours**

**Mode of Evaluation:**

| Evaluation Scheme | | | |
|---|---|---|---|
| MSE | CA | ESE | Total Marks |
| - | 50 | - | 50 |
| 50 | | | |